



Real-time wind estimation from the internal sensors of an aircraft using machine learning

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Abstract

A real-time wind velocity vector and parameters estimation and wind model identification approach using a machine learning technique is addressed in this paper. The proposed method uses only the state measurements of an aircraft and does not require control commands, air data systems, or satellite-based data. Small unmanned aerial vehicles (UAVs) can benefit from this method, since it relies solely on measurement results from the common sensors as an attitude and heading reference system. The independence of external sources of information made estimations resistant to intentional errors. This algorithm uses long short-term memory neural networks (LSTM NNs) in a two-step deep learning process involving classification and regression. A classification NN was trained with four different labeled wind models, while individual regression NNs were trained to estimate the velocity vector and parameters of each wind model. The linear acceleration, angular velocity, and Euler angle measurements were used as the inputs of trained networks. The algorithm suggests in its first step identifying the exact wind model, and in its second step estimating the wind velocity vector and parameters using a properly assigned estimation from a trained network. A nonlinear six-degree-of-freedom simulation of straightforward and level turn maneuvers of a fixed-wing UAV in the presence of different wind models served as the dataset in the learning process. Monte Carlo simulations proved the accuracy and rapidity of the proposed algorithm in identifying the wind model and estimating three-dimensional wind velocity vector and parameters.

Keywords Wind estimation · Machine learning · Unmanned aerial vehicle · Long short-term memory neural network

List of symbols

a_x, a_y, a_z	UAV's linear accelerations in the inertial frame	d_x, d_y, d_z	Gust length in the wind frame
$\mathbf{b}_m$	Bias vector in neural network unit	$\mathbf{f}$	Aerodynamic force vector
b	Wing span	f_X, f_Y, f_Z	Aerodynamic forces in the body frame
$\mathbf{c}$	Cell state in neural network	$\mathbf{g}$	Gravitational acceleration vector
$\bar{c}$	Mean aerodynamic chord	$\mathbf{h}$	Output state in neural network
$C_{X,Y,Z,l,m,n}$	Aerodynamic force and moment coefficients	$\mathbf{I}$	Inertial moment matrix
C_L, C_D	Lift and drag coefficients	$I_{xx}, I_{yy}, I_{zz}, I_{xz}$	UAV's inertial moments in the body frame
		K_1, K_2, K_I	Control gains
		L_u, L_v, L_w	Scale of turbulence along the UAV's body axes
		l, m, n	Aerodynamic moments in the body frame
		$\mathbf{m}$	Aerodynamic moment vector
		m	Mass of the UAV
		p, q, r	UAV's angular velocities in the body frame
		p_w, q_w, r_w	Wind's angular velocities
		$q_{0,1,2,3}$	Coordinates of the rotation quaternion
		r_{max}	Radial distance of the maximum horizontal microburst wind
		$\mathbf{R}$	Transformation matrix

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$\mathbf{R}_m$	Recurrent weight vector in neural network unit
S	Reference (wing) area
$\mathbf{s}$	UAV's position vector
T	Total thrust
u_{max}	Magnitude of the maximum horizontal microburst wind
u, v, w	UAV's linear velocities in the body frame
u_w, v_w, w_w	Wind's linear velocities
u_m, v_m, w_m	Gust amplitude in the wind frame
$\mathbf{v}$	UAV's linear velocity vector
V_a	Airspeed
V_{wh_6}	Wind speed at 6 m altitude
$\mathbf{W}_m$	Input weight vector in neural network unit
$\mathbf{x}$	Neural network input
x, y, z	UAV's positions in the inertial frame
x_{mb}, y_{mb}	Position along the first and second axes of the microburst frame
z_{max}	Altitude of the maximum horizontal microburst wind
α	Angle of attack
α_{mb}	Microburst shaping variable
β	Sideslip angle
δ_t	The throttle setting angle
μ	Mean
ξ	Damping ratio
ρ	Air density
$\sigma_u, \sigma_v, \sigma_w$	Turbulence intensity along the UAV's body axes
$\sigma(\bullet)$	Sigmoid function
σ	Standard deviation
ϕ, θ, ψ	UAV's roll, pitch, and yaw angles
Φ_{u_w, v_w, w_w}	PSD of turbulence along the UAV's body axes
ψ_w	Wind direction at 6 m altitude
Ψ	Phase angle of the PSD
ω_n	Natural frequency
ω	Frequency of the PSD
$\boldsymbol{\omega}$	Angular velocity vector
$\boldsymbol{\Omega}$	Skew-symmetric angular velocity

1 Introduction

The wind is one uncertainty that flying vehicle encounters in the natural environment. It can significantly affect the mission and control of aircraft, especially small UAVs. Besides maintaining the safety and stability of the UAV in its desired path, an accurate wind estimation or

measurement would also help improve the control system's performance when faced with unwanted and unmeasured winds. The ground-based equipment like cup anemometers and more sophisticated remote sensing techniques such as SODARs and LiDARs has been used for wind flow measurement or estimation in aviation and meteorology for decades (Antoniou et al. 2007; Buzdugan and Stefan 2020; Lang and McKeogh 2011). These methods are helpful in identifying, measuring, and estimating atmospheric disturbances at low altitudes and in a limited number of ground sites. An accurately measuring and detecting atmospheric disturbances beyond these ranges requires aircraft with appropriate measurement systems. The ability of manned aircraft and large unmanned aerial vehicles to carry relatively heavy and large measuring equipment has made them excellent options for obtaining atmospheric information (Khelif et al. 1999).

Generally, wind occurs at low altitudes relative to the ground, near terrain, mountains, and areas with varying temperatures. Any type of wind can greatly influence a small UAV's flight characteristics because of its low flight altitude, weight, rotational inertia, size, and speed (compared to the wind speed). Wind as an external factor affects the kinematics and dynamics of the UAV. It is obvious that strong winds cause a small UAV to crash, while less powerful winds can affect its flight path. Recent studies and scientific work have demonstrated that small UAVs are increasingly used to estimate or measure the characteristics of the atmosphere due to their low operating costs (Bailey et al. 2020; Elston et al. 2015; Tian et al. 2021). Using UAVs that can hover has been found very effective in assessing vertical wind profiles (Allison et al. 2020). While small fixed-wing UAVs because of their higher speed and endurance have been used for identifying and estimating the type, intensity of the wind (Borup et al. 2019a, b; Nobahari and Sharifi 2020), and wind field estimation (Langelaan et al. 2011) in a large area. Nevertheless, other methods also have been developed to estimate wind speeds in both vertical and cruise flights (Sun et al. 2019). To accurately estimate wind velocity, the algorithm and method should possess three key characteristics: a fast algorithm, a three-dimensional method, and a method that does not rely on unreliable airflow measurements (Hong et al. 2017). There are direct and indirect methods for estimating wind velocity by UAVs. The direct wind estimating methods are based on airflow measuring equipment and employ the wind triangle approach (Rhudy et al. 2017). Figure 1 shows the concept of wind triangulation. Where, $\mathbf{v}_a$ is the airspeed vector, $\mathbf{v}_w$ is the wind velocity vector, and $\mathbf{v}$ is the linear velocity vector of the UAV. By defining α and β as the angle of attack and sideslip angle, respectively, these vectors can be expressed in the body coordinate system. Therefore, by measuring the triple air parameters

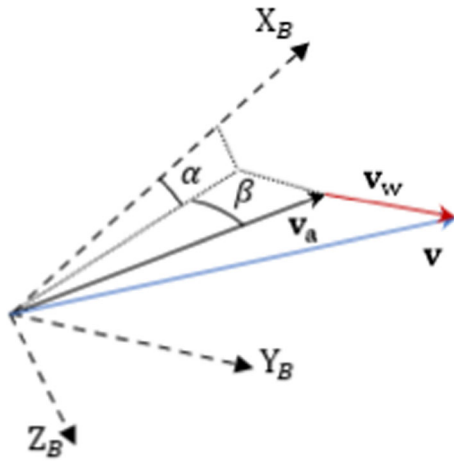


Fig. 1 Wind triangle w.r.t the body coordinate system

(v_a, α, β) , it is possible to calculate the relative velocity of the UAV w.r.t the wind. Using direct wind estimating methods is difficult without information about triple air parameters. Therefore, the UAV must provide with an Air Data System (ADS). Although most fixed-wing aircraft use pitot tubes for measuring the triple air parameters, some small fixed-wing UAVs due to weight and cost limitations cannot use ADS. Even large aircraft with ADS (because of the probability of encountering ADS faults as a frequent incident in aviation accidents (Lu et al. 2016)) use analytical redundancy systems (Lie and Gebre-Egziabher 2013) and indirect approaches to estimate wind velocity and characteristics (Allison et al. 2020) or correct the ADS bias (Borup et al. 2019a, b).

Besides the above reasons, ADS has almost a low-frequency response compared to other measured data from the internal sensors of an aircraft. In some hazardous flights like flying near terrain or mountain, in presence of different wind with rapid pressure change, the flight management system of an autonomous UAV (without weather radar) needs a fast-estimating method to recognize hazardous wind and prepare fast terrain avoidance command. It is clear that reliance on the outputs of the ADS and pitot tubes is not reliable under these conditions.

The indirect wind estimation methods use conventional navigation sensors of the aircraft to estimate the wind. In recent years, several proposals have been made for indirect wind estimation methods, predominantly model-based approaches that necessitate inputs and information from the aircraft controller (Youn et al. 2020). These approaches are usually developed by utilizing the UAV's dynamic model and incorporating the influence of wind on the UAV's movement. The Kalman filter-based methods are the most commonly used method for estimating wind velocity and parameters among all indirect model-based wind

estimation approaches. (Borup et al. 2019a, b; Grillo and Montano 2019; Tian and Chao 2018).

Another effective indirect method for wind estimation problem is the recent application of data-driven and machine learning approaches. Machine learning methods can find relationships between inputs and outputs without relying on problem models. Deep learning as a part of machine learning techniques has attracted much attention for wind prediction and wind speed estimation problems (Devi et al. 2020; Huang et al. 2022; Shin et al. 2021; Vinothkumar and Deeba 2020). Lim et al. (2021) proposed a novel synthetic airspeed estimation method that leverages deep learning and an unscented Kalman filter with data obtained from an inertial measurement unit (IMU), elevator control input, and airflow angles. The performance of LSTM NN for wind speed forecasting is investigated in (Bali et al. 2019; Elsaraiti and Merabet 2021; He et al. 2023; Kim and Lee 2021; Xie et al. 2021), and comparison of the results has shown the superiority of the LSTM NN compared to other neural network algorithms. Allison et al. (2019) and Crowe et al. (2020) have shown that LSTM NN can estimate the wind speed using conventional sensors on a quadrotor, without information about the control system. The input data in the wind estimation method presented by Allison et al. is only the position and Euler angles of a quadrotor in hover flight. Generally, information about the position of the UAV will be achievable in two ways. (1) Using external sources such as GPS. (2) Measurement of position using the INS. Because of insufficient accuracy, the separate INS is no longer of much use in today's systems, and the majority use a combined inertial-satellite system to determine the position (Titterton et al. 2004). There is no doubt that GPS has broad coverage worldwide, but this system is prone to error due to jamming, spoofing, or denying signals in some areas (Lu et al. 2022). Therefore, using the position of UAV and other data that receive from GPS as inputs for an estimation algorithm will reduce the estimation resistance to possible intentional errors.

The wind triangle approach is only reliable for steady state measurements (Allison et al. 2019; Neumann and Bartholmai 2015; Van den Kroonenberg et al. 2008). The Model-based wind estimation methods, despite being widely used in wind estimation problem, also have some shortcomings. The model-based techniques usually use several simplifications to model complex events in presence of nonlinearities pose by intensive continuous or discrete gusts. They do not consider all uncertainty in the parameters of the model or considered linear approximations of nonlinear dynamics (González-Rocha et al. 2019). Where, the unknown functions are not well captured by the developed models and the model cannot include all aspects of the problem (Shamshirband et al. 2019). These simplification limits some model-based wind estimation

approaches to only work in certain situations and apply for only not aggressive flight conditions. Furthermore, the extended nonlinear model-based wind estimation methods rely on the either detailed knowledge of the dynamics and controller of the aircraft (Borup et al. 2016; Liu and Chen 2016; Nobahari and Sharifi 2020), where these methods are not applicable in the case of using most commercial UAVs that do not provide this information for all users. The NNs are effective at approximation of complicated, non-linear, or unknown functions that could not well capture by model-based wind estimation methods (Zimmerman et al. 2022a, b). The wind estimation methods rely on NNs can operate effectively with data related to dynamic flight phases and unsteady condition that imposes in presence of the turbulent wind (Zimmerman et al. 2022a, b). Therefore, the malfunction of the model-based wind estimation techniques and wind triangle estimation methods can compensate with applying NNs for different aspect of wind estimation approaches.

The main contribution of this paper is to develop a fast wind speed and parameters estimation method in the location of a fixed-wing UAV. To this end an indirect wind estimating approach proposed in the framework of using LSTM NNs independent of outputs of the external navigation and air data systems of the UAV that are prone to faults or intentional errors. This work uses the axial acceleration of the UAV that is more affected by minor external wind disturbances than the position of the UAV that used in (Allison et al. 2020). The proposed approach is only relied on the outputs of the internal sensors included linear acceleration, angular velocity, and Euler angle of a fixed-wing UAV. Here, besides using the machine learning method to estimate wind velocity, identifying the model and estimating the wind parameters is also presented. The presented algorithm comprises two phases of classification and estimation, which identify the wind model in the first step and estimate the three-dimensional velocity vector and parameters of the wind in the second step without requiring extra instrumentation, GPS signals, knowledge of the controller or dynamic models of the UAV. The statistical analysis of the results (Monte Carlo tests) obtained for several wind models demonstrated the performance of the proposed method.

The rest of this paper is organized as follows: Sect. 2 develops wind estimation formulation based on incorporating wind effect in aircraft equations of motion. Section 3 introduces wind mathematical models of the winds and Sect. 4 presents the wind estimation problem in the framework of LSTM NNs. Section 5 describes the learning process of wind identification and estimation algorithm. In Sects. 6 and 7, the proposed algorithm is verified and compared to another state-of-the-art approach through simulations and tests in different random conditions.

Finally, Sect. 8 presents a discussion and conclusion about the results.

2 Wind estimation formulation

There are two approaches for incorporating wind velocity into the equations of motion of an aircraft. The first approach considers the wind components as inputs in the aircraft's state-space equations, in addition to the control inputs (McClelland and Woolsey 2020; Nelson 1998). For example, in (Wenz and Johansen 2019), the developed state-space model divides the wind speed into steady and turbulent parts. Meanwhile, with the slowly time-varying assumption of the steady wind part, the Dryden wind model, as a conventional perturbation model, is used for the turbulent component. The second approach considers the effect of wind and perturbations on the airspeed, aerodynamic forces and moments of the aircraft. In the recent approach wind velocity components can be included in the equations of motion of aircraft by defining the relative velocity of the aircraft to the wind (Lungu 2019; Nobahari and Sharifi 2020; Pourmoradi et al. 2021). Our paper uses the second approach to include wind velocity into the equations of motion. In this regard, the translational and rotational six-degree-of-freedom (6DOF) equations of motion of a UAV are developed in Appendix A. According to Eqs. (A.1) to (A.20), the airspeed directly affects the dynamic model of the UAV, and the surrounding air flows around the UAV influence the aerodynamic parameters. As a result, it can be stated that states $u, v, w, p, q, r, \phi, \theta$ and ψ in the 6DOF model of the UAV indirectly are functions of airspeed.

The proposed method in (Allison et al. 2019, 2020) for wind estimation is developed based on the translational equations of motion and the outputs of the navigation system of a quadrotor in the hover mode. In this section, we developed this approach for a fixed-wing UAV in the cruise flight condition. It proves that the equations of motion of a fixed-wing UAV also can form $V_w = f(\cdot)$ to participate in the online wind estimation problem by directly using the outputs of the onboard sensors of the UAV. In this regard, the UAV translational equations of motion (A.1) can be written as:

$$\ddot{x} = vr - wq - g\sin\theta + \frac{T}{m}\delta_t + V_a^2 \frac{\rho S}{2m} C_x \quad (1)$$

$$\ddot{y} = wp - ur + g\cos\theta\sin\phi + V_a^2 \frac{\rho S}{2m} C_y \quad (2)$$

$$\ddot{z} = uq - vp + g\cos\theta\cos\phi + V_a^2 \frac{\rho S}{2m} C_z \quad (3)$$

The Eqs. (1) to (3) discretize as:

$$\begin{aligned}\ddot{x}(k) &\approx \ddot{x}(k-1) + \Delta t \ddot{x}(k-1) = \ddot{x}(k-1) + \Delta t [v(k-1)r(k-1) \\ &\quad - w(k-1)q(k-1) - g \sin \theta(k-1) + \frac{T}{m} \delta_t + V_a^2(k-1) \frac{\rho S}{2m} C_X] \end{aligned} \quad (4)$$

$$\begin{aligned}\ddot{y}(k) &\approx \ddot{y}(k-1) + \Delta t \ddot{y}(k-1) = \ddot{y}(k-1) + \Delta t [w(k-1)p(k-1) \\ &\quad - u(k-1)r(k-1) + g \cos \theta(k-1) \sin \phi(k-1) + V_a^2(k-1) \frac{\rho S}{2m} C_Y] \end{aligned} \quad (5)$$

$$\begin{aligned}\ddot{z}(k) &\approx \ddot{z}(k-1) + \Delta t \ddot{z}(k-1) = \ddot{z}(k-1) + \Delta t [u(k-1)q(k-1) \\ &\quad - v(k-1)p(k-1) + g \cos \theta(k-1) \cos \phi(k-1) + V_a^2(k-1) \frac{\rho S}{2m} C_Z] \end{aligned} \quad (6)$$

The Eqs. (4), (5) and (6) can be rewritten as follows:

$$\begin{aligned}V_a^2(k-1) &= C_X^{-1} \frac{2m}{\rho S} \times \left[\frac{\dot{x}(k) - \dot{x}(k-1)}{\Delta t} - v(k-1)r(k-1) \right. \\ &\quad \left. + w(k-1)q(k-1) + g \sin \theta(k-1) - \frac{T}{m} \delta_t(k-1) \right] \end{aligned} \quad (7)$$

$$\begin{aligned}V_a^2(k-1) &= C_Y^{-1} \frac{2m}{\rho S} \times \left[\frac{\dot{y}(k) - \dot{y}(k-1)}{\Delta t} - w(k-1)p(k-1) \right. \\ &\quad \left. + u(k-1)r(k-1) - g \cos \theta(k-1) \sin \phi(k-1) \right] \end{aligned} \quad (8)$$

$$\begin{aligned}V_a^2(k-1) &= C_Z^{-1} \frac{2m}{\rho S} \times \left[\frac{\dot{z}(k) - \dot{z}(k-1)}{\Delta t} - u(k-1)q(k-1) \right. \\ &\quad \left. + v(k-1)p(k-1) - g \cos \theta(k-1) \cos \phi(k-1) \right] \end{aligned} \quad (9)$$

According to the above equations, airspeed is a function of states $u, v, w, p, q, r, \phi, \theta$ and other constant parameters (assuming that the dimensionless aerodynamic coefficients are constant parameters). The Eqs. (7), (8) and (9) can be simplified by defining B_1, B_2 and B_3 as constant parameters including dimensionless aerodynamic coefficients, mass, gravity acceleration, reference area (wing), trust, and air density as:

$$\begin{aligned}V_a^2(k-1) &= f_1(u(k), u(k-1), v(k-1), w(k-1), q(k-1), r(k-1), \\ &\quad \theta(k-1), \delta_t(k-1), B_1) \end{aligned} \quad (10)$$

$$\begin{aligned}V_a^2(k-1) &= f_2(v(k), v(k-1), u(k-1), w(k-1), p(k-1), r(k-1), \\ &\quad \theta(k-1), \phi(k-1), B_2) \end{aligned} \quad (11)$$

$$\begin{aligned}V_a^2(k-1) &= f_3(w(k), w(k-1), u(k-1), v(k-1), p(k-1), q(k-1), \\ &\quad \theta(k-1), \phi(k-1), B_3) \end{aligned} \quad (12)$$

The airspeed can be independent of the throttle angle $\delta_t(k-1)$ in Eq. (10), since the throttle angle can be calculated from the implemented control law depending on the flight trim condition (Allison et al. 2020). Based on the developed Eqs. (A.1) to (A.5), the airspeed can be considered as a nonlinear function of the states of the UAV and a constant parameter B (including B_1, B_2 and B_3), as follows:

$$\begin{aligned}V_a(k-1) &= f_4([\mathbf{v}_B^E(k)]^B, [\mathbf{v}_B^E(k-1)]^B, [\boldsymbol{\omega}^{BE}(k-1)]^B, \theta(k-1), \\ &\quad \phi(k-1), B) \end{aligned} \quad (13)$$

We can state the linear velocity of the UAV w.r.t Earth ($[\mathbf{v}_B^E]^B$) at discrete times (k) and $(k-1)$ is a good approximation of the acceleration at time (k) , in the body frame. Therefore, Eq. (13) can be rewritten as:

$$V_a(k-1) = f_4([\mathbf{v}_B^E(k)]^B, [\boldsymbol{\omega}^{BE}(k-1)]^B, \theta(k-1), \phi(k-1), B) \quad (14)$$

Based on the wind triangle relationship (Rhudy et al. 2017) (wind velocity is the difference between the ground velocity and the airspeed) and Eq. (14), it is possible to show that the wind velocity can be learned (Allison et al. 2020) as a nonlinear function of the linear acceleration and angular velocity of the UAV w.r.t Earth, bank angle, pitch angle, and constant B . Since the UAV also flies with a non-zero yaw angle in some missions, this angle must add to the mentioned parameters. Finally, if we consider that every state of a UAV (with a continuous dynamic and an effective controller) is affected by the previous n time steps of that state, the wind velocity be depending on a sequence of these data and becomes:

$$\begin{aligned}V_w(k-1) &= f_5([\mathbf{v}_B^E(k)]^B, [\mathbf{v}_B^E(k-1)]^B, \dots, [\mathbf{v}_B^E(k-n)]^B, [\boldsymbol{\omega}^{BE}(k)]^B, \\ &\quad \dots, [\boldsymbol{\omega}^{BE}(k-n)]^B, \\ &\quad \phi(k), \dots, \phi(k-n), \theta(k), \dots, \theta(k-n), \psi(k), \dots, \psi(k-n), B) \end{aligned} \quad (15)$$

3 Mathematical modeling of the atmospheric disturbances

The dynamic state variables and initial flight trajectory of a UAV are affected differently by various wind conditions, like stochastic or semi deterministic winds. If the wind intensity (as an external disturbance) exceeds the control capabilities of the controller, it could cause deviations, drifts, or delays in the desired trajectory of the UAV. Meanwhile, in the worst conditions, strong winds can destroy the UAV. There are various representations of

wind modeling, such as representations in the time domain (Shamshad et al. 2005), frequency domain (Wright and Cooper 2008), and spatial descriptions (Zhenxing and Hongbin 2009). Therefore, we introduce some models of common winds that can affect small UAVs at low altitudes, such as constant wind, turbulence (Dryden model), 1-cosine wind, and microburst. These are some of common wind models that are used to model and simulate the natural environment in wind estimation problems (Allison et al. 2020; Nobahari and Sharifi 2020).

3.1 Constant wind model

The constant wind is considered as an average of the wind speed and direction over a relatively short period. For this type of the wind, the following model can be considered:

$$u_w = \text{Constant} \quad (16)$$

$$v_w = \text{Constant} \quad (17)$$

$$w_w = \text{Constant} \quad (18)$$

where u_w , v_w and w_w are the wind speed in the north, east and down direction in a local frame (NED), respectively.

3.2 Turbulence model

When the distance scale is divided by the forward speed of an aircraft, the gust profile becomes a gust velocity time history. A gust profile is usually a continuous and irregular profile that is known as turbulence (Fonte et al. 2015). In aerospace applications, turbulence can be modeled as a superposition of an infinite number of sinusoidal components of individual gusts and their Power Spectral Density (PSD) as:

$$V_{w_i}(t) = \sum_{k=1}^{\infty} \sqrt{\Phi(\omega_k) \Delta \omega} \cos(\omega_k t + \Psi_k) \quad (19)$$

where $\Phi(\omega_k)$ is the PSD of each component (gust) with the frequency of ω_k and phase angle Ψ_k . The von Kármán and Dryden models are the most famous turbulence models developed based on the PSD concept. Here, we use the Dryden model to represent turbulence, while the PSD for this model is expressed as Eqs. (20) to (22) along the UAV's body axes (Madden 2018).

$$\Phi_{u_w}(\omega) = \sigma_u^2 \frac{2L_u}{\pi V_a} \frac{1}{\left[1 + \left(L_u \frac{\omega}{V_a}\right)^2\right]^2} \quad (20)$$

$$\Phi_{v_w}(\omega) = \sigma_v^2 \frac{L_v}{\pi V_a} \frac{1 + 3\left(L_v \frac{\omega}{V_a}\right)^2}{\left[1 + \left(L_v \frac{\omega}{V_a}\right)^2\right]^2} \quad (21)$$

$$\Phi_{w_w}(\omega) = \sigma_w^2 \frac{L_w}{\pi V_a} \frac{1 + 3\left(L_w \frac{\omega}{V_a}\right)^2}{\left[1 + \left(L_w \frac{\omega}{V_a}\right)^2\right]^2} \quad (22)$$

where σ_i and L_i (for $i = u, v, w$) are turbulence intensity and scale of turbulence, respectively. Also V_a is expressed as true airspeed. The Dryden wind model is simulated using the MATLAB continuous Dryden wind block. The developed algorithm in this block implemented the PSD through forming filters on band-limited white noise. The main inputs for this block are V_{w_6} as wind speed at 6 m altitude (defines the low-altitude intensity), and ψ_w as wind direction at 6 m altitude (clockwise from the north).

3.3 1-cosine wind model

A series of individual gusts can create a continuous turbulence profile and an idealized individual gust or discrete gust is called the 1-cosine gust model (as shown in Fig. 2). Equation (23) provides a mathematical model for 1-cosine wind speed in the body coordinate system (Nobahari and Sharifi 2020).

$$v_{1-\text{cosine}} = \begin{cases} \frac{v_m}{2} \left(1 - \cos\left(\frac{2\pi x_g}{d_m}\right)\right) & \text{if } 0 < x_g < d_m \\ 0 & \text{otherwise} \end{cases} \quad (23)$$

where v_m , d_m and x_g are gust amplitude, gust length and distance traveled, respectively.

3.4 Microburst wind model

Wind shear is one of the most dangerous environmental factors for the safety of an unmanned aerial vehicle. The characteristic of this type of the wind is the rapid change of its velocity at a relatively short distance. This phenomenon is evident in mountainous and urban environments with tall

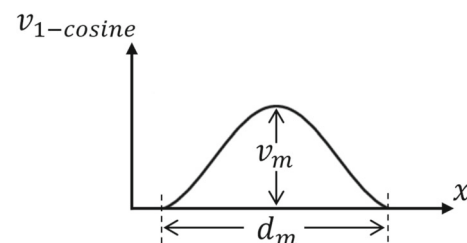


Fig. 2 Displaying of a 1-cosine pulse

buildings. In most cases, the dangerous type of this wind is associated with convective outflows of thunderstorms at low altitudes, known as a microburst (Vicroy 1991). As schematically illustrated in Fig. 3, the microburst is a strong cylindrical localized downdraft that shifts rapidly from a headwind to a tailwind when it impacts the ground (Vicroy 1994). In the absence of a proper flight controller, this atmospheric disturbance can seriously endanger the flight safety of any aircraft, especially small UAVs. Based on the coordinate system defined in Fig. 4 and assuming that the center of the microburst field lies in the center of the frame, it is possible to derive the following mathematical expression to calculate the velocity of the microburst:

$$u_{mb} = \frac{\lambda x_{mb}}{2} \left[\exp\left(c_1 \frac{h}{z_{max}}\right) - \exp\left(c_2 \frac{h}{z_{max}}\right) \right] \exp\left(\frac{2 - x_{mb}^{2\alpha_{mb}} / r_{max}^{2\alpha_{mb}}}{2\alpha}\right) \quad (24)$$

$$v_{mb} = \frac{\lambda y_{mb}}{2} \left[\exp\left(c_1 \frac{h}{z_{max}}\right) - \exp\left(c_2 \frac{h}{z_{max}}\right) \right] \exp\left(\frac{2 - (x_{mb}^2 + y_{mb}^2)^{\alpha_{mb}} / r_{max}^{2\alpha_{mb}}}{2\alpha}\right) \quad (25)$$

$$w_{mb} = -\lambda \left(1 - \frac{x_{mb}^{2\alpha_{mb}}}{2r_{max}^{2\alpha_{mb}}} \right) \exp\left(\frac{2 - x_{mb}^{2\alpha_{mb}} / r_{max}^{2\alpha_{mb}}}{2\alpha}\right) \left\{ \frac{z_{max}}{c_1} \left[\exp\left(c_1 \frac{h}{z_{max}}\right) - 1 \right] - \frac{z_{max}}{c_2} \left[\exp\left(c_2 \frac{h}{z_{max}}\right) - 1 \right] \right\} \quad (26)$$

where r_{max} is radial distance of the maximum horizontal wind, z_{max} is altitude of the maximum horizontal wind, u_{max} is magnitude of the maximum horizontal wind, x_{mb} and y_{mb} are the position along the first and second axes of the microburst frame, respectively. Additionally, α_{mb} as a shaping variable is equal to 2 (Pourtakdoust et al. 2011), and c_1, c_2 are equal to -0.15 and -3.2175 , respectively (Vicroy 1994). The parameter λ as a scale factor expressed as:

$$\lambda = \frac{2u_{max}}{r_{max}[\exp(c_1) - \exp(c_2)] \exp\left(\frac{1}{2\alpha_{mb}}\right)} \quad (27)$$

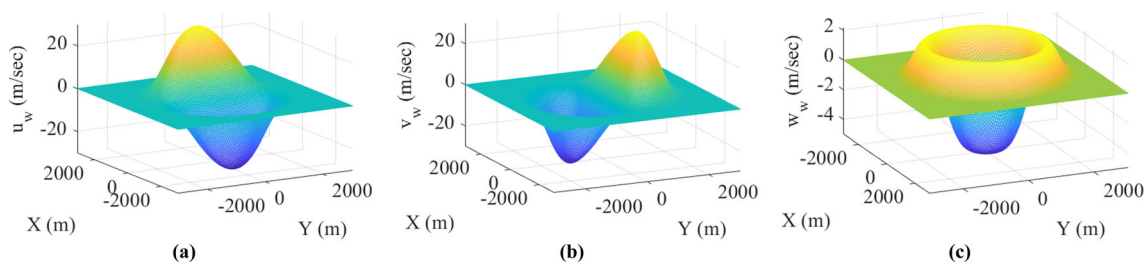


Fig. 3 Microburst velocity field structure

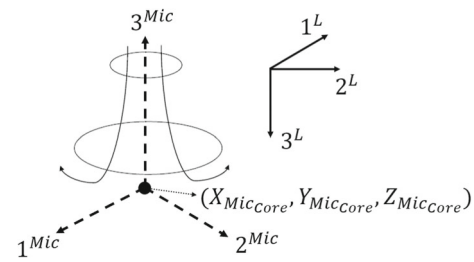


Fig. 4 Definition of the coordinate system in the microburst model related to a local frame system

4 LSTM recurrent neural networks

The states of the UAV are sequence data that we used for the classification of wind models and estimation of the wind velocity vector and wind's parameters. In the last decade, the traditional neural network was unsuccessful at dealing with the sequences data. Recurrent neural networks (RNNs), particularly back-propagation through time (BPTT) algorithms, are considered in time-series predictions and processing sequence data due to their ability to remember previous information and apply it to the current prediction calculation (Medsker and Jain 2001; Xie et al. 2021). The RNNs have a problem learning long-term dependencies and may encounter gradient disappearance or gradient explosion (Smagulova and James 2019). LSTM is an advanced type of RNN that was introduced by Hochreiter et al. (Hochreiter and Schmidhuber 1997). This type of RNNs uses a memory cell and multiple control gates to combine short-term and long-term memory to process sequential and temporal data. LSTM has the same structure as standard RNNs, but differs in cell composition. It introduced a forget gate into the standard recurrent sigma cell of the RNNs to address the difficulty encountered by RNNs (Beu and Landolfo 2023). Figure 5 shows the structure of LSTM neuron. Every LSTM neuron includes the input, forget, and output gates. The input gate determines and updates the new information that will be stored in the cell. The output gate is responsible for determining which part of the current cell state goes to the output. The forget gate identifies unwanted information and throws it from the cell state by running a sigmoid layer. All the

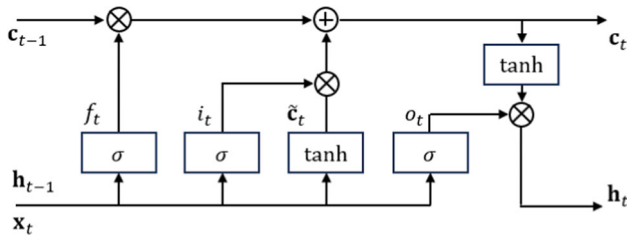


Fig. 5 Basic structure of the LSTM neuron

introduced gates use separate sigmoid layers (Wu et al. 2020; Xiang et al. 2020). By defining the input weight vector as $\mathbf{W}_m = [W_i \ W_f \ W_g \ W_o]^T$, the recurrent weight vector as $\mathbf{R}_m = [R_i \ R_f \ R_g \ R_o]^T$, and the bias vector as $\mathbf{b}_m = [b_i \ b_f \ b_g \ b_o]^T$, the LSTM layers can be formulated as Eqs. (28) to (33) (Ma et al. 2015). The input, output, and forget gates are denoted as i , o , and f , respectively.

$$i_t = \sigma(W_i \mathbf{x}_t + R_i \mathbf{h}_{t-1} + b_i) \quad (28)$$

$$f_t = \sigma(W_f \mathbf{x}_t + R_f \mathbf{h}_{t-1} + b_f) \quad (29)$$

$$\tilde{c}_t = \sigma(W_g \mathbf{x}_t + R_g \mathbf{h}_{t-1} + b_g) \quad (30)$$

$$\mathbf{c}_t = f_t \odot \mathbf{c}_{t-1} + i_t \odot \tilde{c}_t \quad (31)$$

$$o_t = \sigma(W_o \mathbf{x}_t + R_o \mathbf{h}_{t-1} + b_o) \quad (32)$$

$$\mathbf{h}_t = o_t \odot \sigma(\mathbf{c}_t) \quad (33)$$

where $\mathbf{h}_t$ is known as output or hidden state vector of the network. A cell state vector, $\mathbf{c}_t$, is calculated according to the network input in the time step, $\mathbf{x}_t$, the output state and the cell state in time step $t-1$, $(\mathbf{h}_{t-1}, \mathbf{c}_{t-1})$. The state activation functions, $\sigma(\bullet)$, for each gate in this network, are calculated using a sigmoid function as Eq. (34). The sign $\odot$ represents Hadamard product.

$$\sigma(\mathbf{x}) = (1 + e^{-\mathbf{x}})^{-1} \quad (34)$$

The input vector of the LSTM network for wind estimation problem is considered as Eq. (35). The input vector contains nine states $(a_x, a_y, a_z, p, q, r, \phi, \theta, \psi)$. It is possible to measure these states using a simple sensor like an attitude and heading reference system (AHRS) (a common sensor in all aerial vehicles).

$$\mathbf{x}_t = \left([\dot{\mathbf{v}}_B^E(k)]^B, [\dot{\mathbf{v}}_B^E(k-1)]^B, \dots, [\dot{\mathbf{v}}_B^E(k-n)]^B, [\boldsymbol{\omega}^{BE}(k)]^B, \dots, [\boldsymbol{\omega}^{BE}(k-n)]^B, \phi(k), \dots, \phi(k-n), \theta(k), \dots, \theta(k-n), \psi(k), \dots, \psi(k-n) \right) \quad (35)$$

Equations (1) to (35) described the wind velocity estimation problem in the framework of an LSTM network. This approach uses the input (35) as states of a fixed-wing UAV that is sensed by an attitude and heading reference system (Titterton et al. 2004).

The proposed wind estimation and identification algorithm can be divided into two stages: (1) wind model identification, (2) wind velocity vector estimation and wind model parameter estimation. In the first step, a trained classification LSTM network with inputs (35) is used to identify the wind model. Following identifying the wind model and allocating the appropriately trained estimation LSTM network, the wind velocity vector and wind model parameters (such as amplitude and intensity and etc.) is estimated using the inputs (35). It should note that the wind velocity is a function of wind model parameters, consequently, the $\mathbf{x}_t$ is also a function of these parameters. So, with the same logic, we can estimate the wind parameters based on the input $\mathbf{x}_t$. Figure 6 shows the framework of the proposed online wind identification and estimation algorithm.

The first step in the online wind estimation process in this paper is the identification and classification of the wind types. In this regard, the LSTM network for sequence-to-label classification, with the structure shown in Fig. 7a, is used for classification neural network. This NN classified the wind models based on labels 1, 2, 3, and 4, respectively, for constant wind, turbulence, 1-cosine wind, and microburst wind. In the next step, the LSTM network for sequence-to-sequence regression (for estimation of the wind velocity vector and parameters), with the structure shown in Fig. 7b, is used for each type of wind. As shown in Fig. 7, a deep LSTM network is achieved by using two LSTM layers, while over-fitting is prevented by adding a dropout layer after each LSTM layer. Test results showed that the trained network with two layers of LSTM gave better accuracy compared to one layer.

5 The learning process of LSTM NNs

A closed-loop nonlinear 6DOF simulator of the motion of a small fixed-wing UAV with the specifications mentioned in Appendix B is developed in Simulink/MATLAB environment. The simulated UAV, because of its low weight and high aspect ratio, is very sensitive to small atmospheric disturbances. This condition helps the proposed algorithm to identify the smallest atmospheric disturbances with more accuracy.

Various controllers can be used in the autopilot of a fixed-wing UAV, depending on the desired performance. This paper proposes to use the UAV acceleration state variables as one-third of the input parameters in the wind estimation algorithm. Therefore, using an acceleration controller to generate the acceleration command for tracking the predefined trajectory seems more helpful in the wind estimation algorithm. In this regard, we used the acceleration autopilot proposed in (Esmailifar and Saghafi

Fig. 6 The framework of online wind identification and estimation algorithm

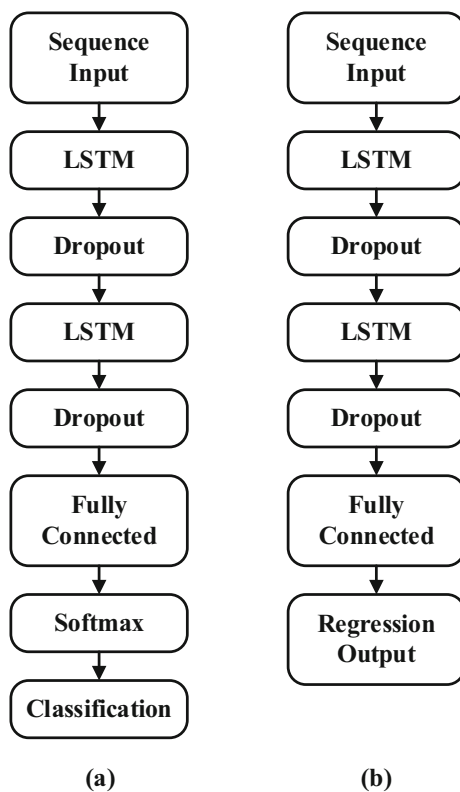
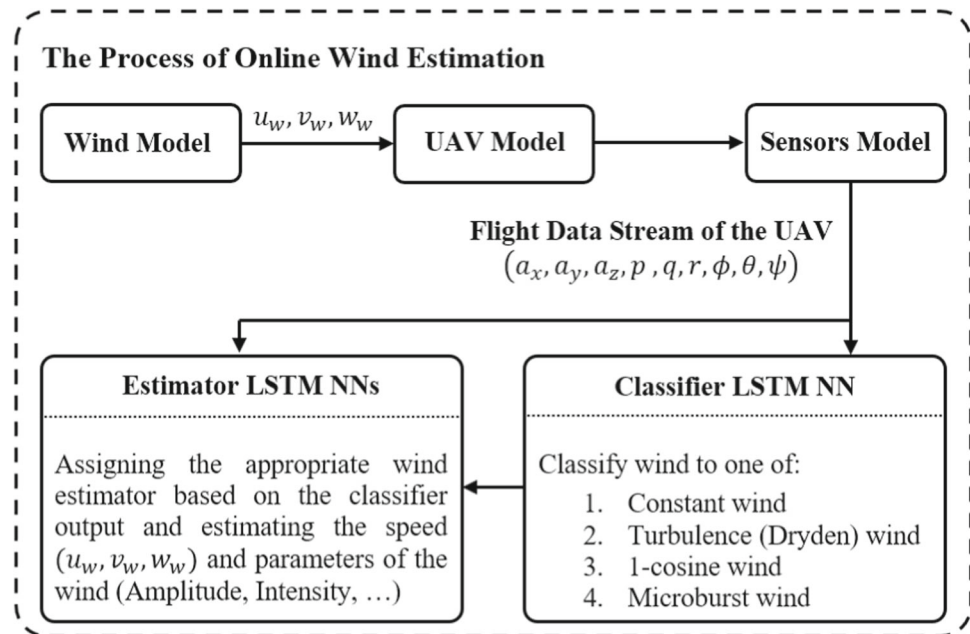


Fig. 7 The architecture of a LSTM NN for **a** classification and **b** regression

2017) for the close loop nonlinear 6DOF simulation. This controller combined state feedback with an integrator in each channel of the control system. The general structure of the bank, normal and lateral acceleration control system,

is shown in Fig. 8. Moreover, the reference commands, feedback signals, and control gains of all autopilot channels introduce in Appendix B. We used a linear guidance method (Askari et al. 2016) to generate the reference acceleration and bank angle commands to track a predefined trajectory in all simulations.

The architecture of the offline training dataset generation process by a flight simulator is depicted in Fig. 9. The wind models, nonlinear 6DOF model of the UAV and sensors models are simulated in the Simulink/MATLAB environment. The developed simulator provides the required data for training and testing of the LSTM NNs. In this regard, the nonlinear 6DOF simulation software ran 1000 times separately with each type of wind model (in total, 4000 times). The simulation time was between 4 and 25 s per run (depending on the wind model), while all simulations used a fixed time step of 0.01 s. We used a piecewise signal for training the constant wind model, where the wind vector magnitude changed randomly every 5–15 s. The wind parameters for other wind models were constant during each run. A variety of level turns and

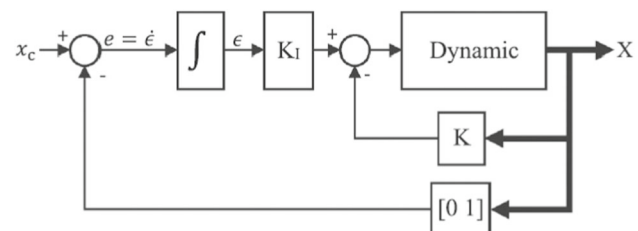
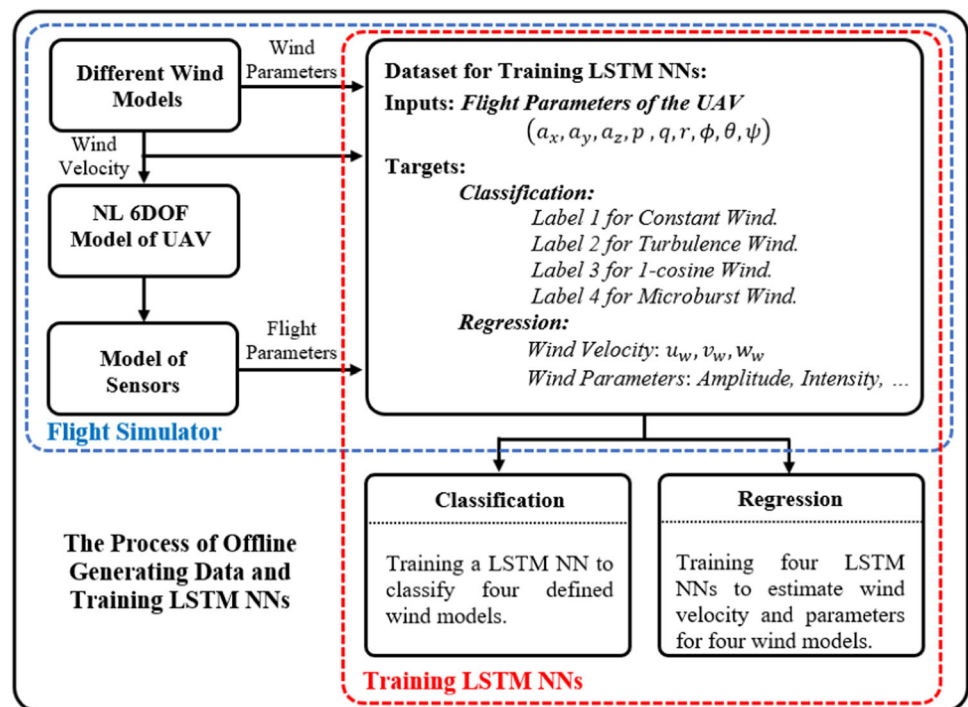


Fig. 8 The acceleration autopilot for close loop simulator of the UAV (Esmailifar and Saghafi 2017)

Fig. 9 The architecture of training and generating training data set based on the 6DOF model of the UAV



straightforward maneuvers at different altitudes are simulated to create diversity in the data used for neural networks training. Generally, these simulations have extracted at least 4000 s of data for each wind model in trim flight conditions. Each simulation run is considered a random selection of the parameters for each wind model. These parameters are listed in Table 1, along with their allowable ranges. According to Fig. 10, the normal distribution of the selected random data can be seen in the plotted probability density function (PDF) related to each parameter. In addition, the mean (μ) and standard deviation (σ) of the selected random data are presented in Table 1.

The architecture of the offline training process of LSTM NNs is illustrated in Fig. 9. The nonlinear 6DOF simulator simulates the flight of UAV in the presence of different winds (in MATLAB environment) and provides comprehensive data, which is used as inputs and targets for the LSTM NNs. In this work, we used the framework supplied by the Deep Learning Toolbox of MATLAB (DLTM) for LSTM NNs in classification and regression moods. We implemented the proposed LSTM NN structures indicated in Fig. 7a and b for classification and regression tasks, respectively.

Table 1 The variations interval and properties of the selected wind model parameters

Wind model	Parameter	Unit	variations interval	The properties of the selected random data	
				μ	σ
Constant	u	m/s	− 10 to 10	− 1.20	3.69
	v	m/s	− 10 to 10	− 0.76	4.21
Dryden	V_{whg}	m/s	1 to 10	4.54	2.88
	ψ_w	deg	− 90 to 90	3.68	51.85
1-cosine	$u_m = v_m$	m/s	1 to 10	5.89	2.89
	w_m	m/s	− 10 to 10	− 0.17	5.74
	$d_x = d_y$	m	80 to 120	100.79	11.71
	d_z	m	80 to 120	99.50	11.6
Microburst	u_{max}	m/s	1 to 10	5.37	2.65
	z_{max}	m	100 to 200	149.67	27.72
	r_{max}	m	100 to 1000	550.22	249.23

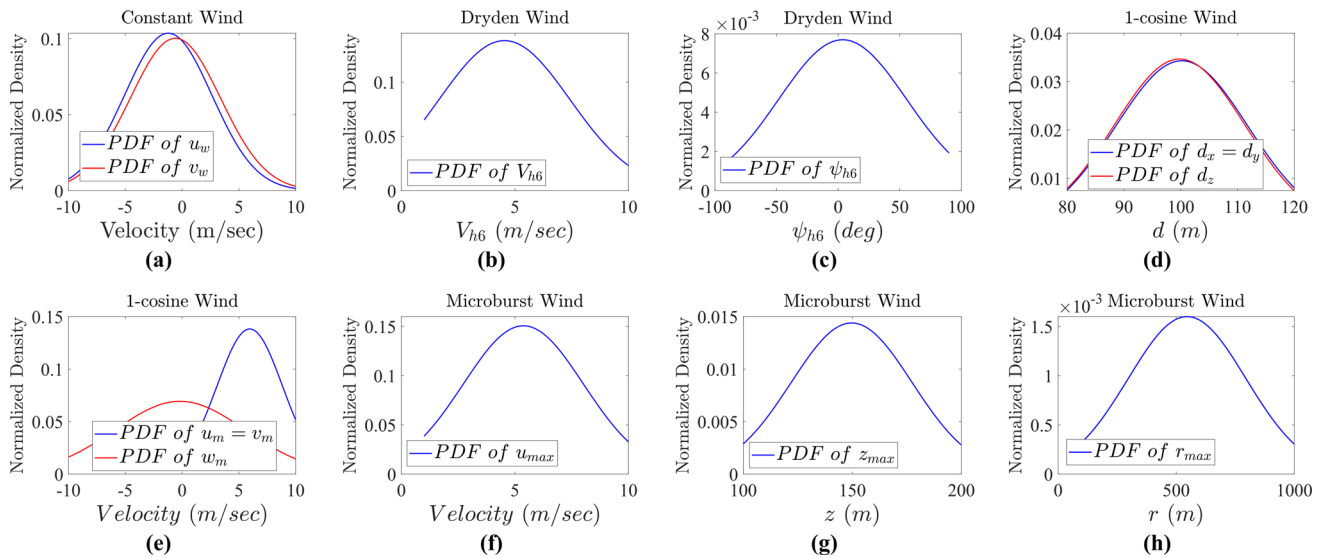


Fig. 10 The normalized PDF of wind model variables: **a** Constant wind velocity. **b** $V_{w_{h6}}$ for Dryden wind. **c** ψ_w for Dryden wind. **d** d for 1-cosine wind. **e** Velocity for 1-cosine wind. **f** u_{max} for Microburst wind. **g** z_{max} for Microburst wind. **h** r_{max} for Microburst wind

The flight parameters contain a sequence of $a_x, a_y, a_z, p, q, r, \phi, \theta, \psi$ is input vector in the training process of all LSTM networks. The labels 1, 2, 3, and 4, respectively, for constant wind, turbulence, 1-cosine wind, and microburst wind used as the target in the training process of classification. The wind velocity (u_w, v_w, w_w) beside the wind parameters (depending on the wind model) are used as the target vectors in the training process of regression networks. The states, wind velocity, and parameters used as inputs and targets in LSTM have different scales. Therefore, a hyperbolic tangent function used in LSTM maps the features to a range $[-1, 1]$. By utilizing a normalization operation represented by Eq. (36), we standardized all inputs and targets, thereby rescaling the data and ensuring that each feature has an equal influence on the ultimate estimation outcomes. Where, μ and σ in Eq. (36) are mean and the standard deviation vectors of the $\mathbf{x}$, respectively.

$$\mathbf{x}_{norm} = \frac{\mathbf{x} - \mu}{\sigma} \quad (36)$$

A neural network algorithm's result relies heavily on the loss function and solver, so selecting the appropriate options for these parameters is vital. In the DLTM, stochastic gradient descent with momentum (SGDM), RMSProp, and Adam (Kingma and Ba 2014) are available as optimizer options. Based on comparing these methods, the Adam optimizer outperformed others in our classifying and estimating wind problem. So, it is used in all developed trained networks.

The hyperparameters are the higher-level parameters that control the training processes and should be set before starting the training of a machine learning model. They can

be adjusted based on properties such as the characteristics of the data and the capacity of the algorithm to learn (Agrawal 2021; Nematzadeh et al. 2022). In this work, we manually search and adjust the hyperparameters of the proposed LSTM NNs. The selected LSTM hyperparameters for all trained networks are illustrated in Table 2. The learning rate is selected equal to 0.001 based on a similar wind estimation problem that was addressed in (Allison et al. 2019). Our investigation shows that although this value increases the convergence time of the training process slightly, it can guarantee that we do not miss any local minimums. To determine appropriate values for the remaining hyperparameters, a trial-and-error method outlined in Appendix C is employed. Where, the authors used their expertise and relevant references (Crowe et al. 2020; Elsaraiti and Merabet 2021; Lim et al. 2021; Zimmerman et al. 2022a, b; Zimmerman et al. 2022a, b) to derive an initial estimate for each parameter and repeated the trial-and-error process to achieve a neural network with the best accuracy for the classification approach and four neural networks with minimum RMSE for the regression approach.

The evaluation criteria for the trained networks differ based on their purpose. For classification, accuracy is deemed suitable while for estimation, the root mean squared error (RMSE) and coefficient of determination or R-squared (R^2) criteria are chosen. The values of accuracy, RMSE and R^2 are calculated based on Eqs. (37), (38) and (39), respectively. Where Y defines the test data, $\hat{Y}$ is the estimated data, $\bar{Y}$ is the mean of the test data, and N is the number of samples.

Table 2 The LSTM parameter settings

Parameter	Classification	Regression			
		Constant	Dryden	1-cosine	Microburst
Learning rate	0.001	0.001	0.001	0.001	0.001
Hidden unit	100	200	200	200	200
Min batch size	25	30	27	27	25
Epochs of training	200	300	250	200	150

$$Accuracy = \frac{\sum_{i=1}^N (Y_i == \hat{Y}_i)}{N} \times 100 \quad (37)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (Y_i - \hat{Y}_i)^2}{N}} \quad (38)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \quad (39)$$

6 Simulation results

After training one neural network for the classification of the wind and four separate neural networks for the estimation of the wind velocity and parameters, we test these trained networks with new test data. The simulated sensors in the 6DOF simulator modeled as zero dynamic with noisy output [as a proposed model by H. Nobahari et al. in (Nobahari and Sharifi 2020)] while utilizing a dynamic filter can remove the sensor errors, except for the noise (measurement noise). It is clear that the accuracy of estimation is decreased compared with clean conditions in the presence of noise. The simulation-based studies showed a small and acceptable deviation for all estimation results in this work. The following subsections present some figures and tables for describing the simulation results.

6.1 Identification of the type of wind

The online wind estimation process illustrated in Fig. 6 involves identifying the type of wind in the first step. It is evident that the specification of all test data cannot be represented in a limited number of pages of this paper. Therefore, we consider a random dataset comprising three different samples of each wind model (twelve samples for all models) to quantitatively investigate the accuracy of the trained classification and regression networks. It should be considered that the parameters for each model are randomly chosen within the allowable range mentioned in Table 1. Table 3 presents the specifications of each test sample. The results of this test data show that the trained

classification LSTM NN could identify each model with the exact corresponding label, where the accuracy equal to 100% is extracted based on Eq. (37) for this data set. Although the high accuracy results are obtained because of the good training of the LSTM NN but the wind models considered here including constant wind, Dryden wind, 1-cosine wind, and microburst wind have different behavioral patterns, and as a result, each one has a particular effect on the states of the UAV.

6.2 The estimation of wind velocity and parameters

In the second step of the online wind estimation process, the wind velocity and parameters are estimated. For this purpose, after determining the type of wind in the first step, by assigning the appropriate wind LSTM NN, the wind velocity and parameters are estimated with the same input $\mathbf{x}_t$. Table 4 provides the estimated parameters for samples A, B, and C of each wind model (introduced in Table 3). Also, Figs. 11, 12, 13, 14, 15, 16 and 17 show the performance of wind velocity estimation method for test sample A in each wind model.

6.2.1 Estimation of constant wind

The LSTM estimation network trained for constant wind is investigated for the data set mentioned in Table 3. Figure 11a–c show estimation results (sample A) for the sequence of wind velocity components compared to their test values. Also, Fig. 10d–f show the error diagram based on the difference between the estimated value and the actual value for each velocity component. In each piece of constant wind, the error value is high in the first few time samples due to sharp transitions in speed, but after a few time samples, the estimation converges to the test value, and the error has been drastically reduced. Figure 12 shows a piece of the wind sequence (1.50 s) separately to enhance the comparability of the results obtained for the constant wind estimation. A rapid convergence of estimated results to real values is observed in less than 0.5 s.

Table 3 The wind specification of the test dataset

Wind model	Classification label	Parameter	Unit	Test data A	Test data B	Test data C
Constant	1	u	m/sec	− 2.4, 3.4, 0	4.4, − 8.5, 1.3	− 5.0, 0, 3.0
		v	m/sec	− 0.5, − 6.2, 0	− 6.8, 0, − 4.0	7.7, 2.6, − 4.0
		w	m/sec	0.0	0.0	0.0
Dryden	2	V_{wh_6}	m/sec	2.1	6.1	− 0.5
		ψ_w	deg	− 14	58	− 75
1-cosine	3	$u_m = v_m$	m/sec	7.7	2.4	8.2
		w_m	m/sec	− 8.8	− 1.7	− 7
		$d_x = d_y$	m	100	100	120
		d_z	m	109.8	90	85
Microburst	4	u_{max}	m/sec	4.2	1.8	2.7
		z_{max}	m	188	177	150
		r_{max}	m	193	739	182

Table 4 The RMSE of wind velocity and parameters estimation for different wind models

Wind model	Parameter	RMSE for samples			Wind model	Parameter	RMSE for samples		
		A	B	C			A	B	C
Constant	u_w	0.21	0.62	0.12	Dryden	u_w	0.087	0.138	0.157
	v_w	0.15	0.50	0.23		v_w	0.043	0.065	0.109
	w_w	0.00	0.00	0.00		w_w	0.020	0.031	0.061
1-cosine	u_w	0.075	0.056	0.194	Microburst	u_w	0.163	0.182	0.252
	v_w	0.051	0.027	0.129		v_w	1e−4	3e−4	6e−4
	w_w	0.031	0.018	0.083		w_w	0.064	0.032	0.071
	$u_m = v_m$	0.718	0.531	0.339		u_{max}	2.2	2.05	0.64
	w_m	0.231	0.115	0.756		z_{max}	2.85	14.57	9.42
	$d_x = d_y$	1.746	1.427	2.346		r_{max}	33.14	16.94	27.25
	d_z	1.953	1.481	3.419		—	—	—	—

6.2.2 Estimation of turbulence (Dryden)

The RMSE of tested LSTM for three Dryden wind samples A, B, and C, with specifications mentioned in Table 3, are shown in Table 4. The performance of the wind estimation algorithm for test sample A of the Dryden wind is shown in the first column of Fig. 13, while the diagram of the estimated values error is presented in the second column of it. It is evident that the error was bounded around small values for all components of estimated wind. Moreover, small RMSE verified the acceptability of the estimation results for the trained network.

6.2.3 Estimation of 1-cosine wind

The performance of the trained estimation LSTM network is tested for three sets of simulation state variables of the UAV in the presence of a 1-cosine wind model. The three-

dimensional parameters of 1-cosine wind are gust amplitude ($u_m = v_m, w_m$) and gust length ($d_x = d_y, d_z$). We tested the 1-cosine trained network for estimating these parameters and the wind velocity in a local frame. The performance of estimation results (for sample A of Table 3) is shown in Fig. 14 and Fig. 15. As shown in Fig. 14, the estimation wind velocity error is bounded around small values across the wind profile. Figure 15 shows the estimation error for gust amplitude and gust length converged to small values, after less than one second. These results indicated the effectiveness of the trained network and small RMSE for this kind of wind.

6.2.4 Estimation of microburst wind

The first column of Figs. 16 and 17 compared the estimated wind velocity and wind parameters u_{max} , z_{max} , and r_{max}

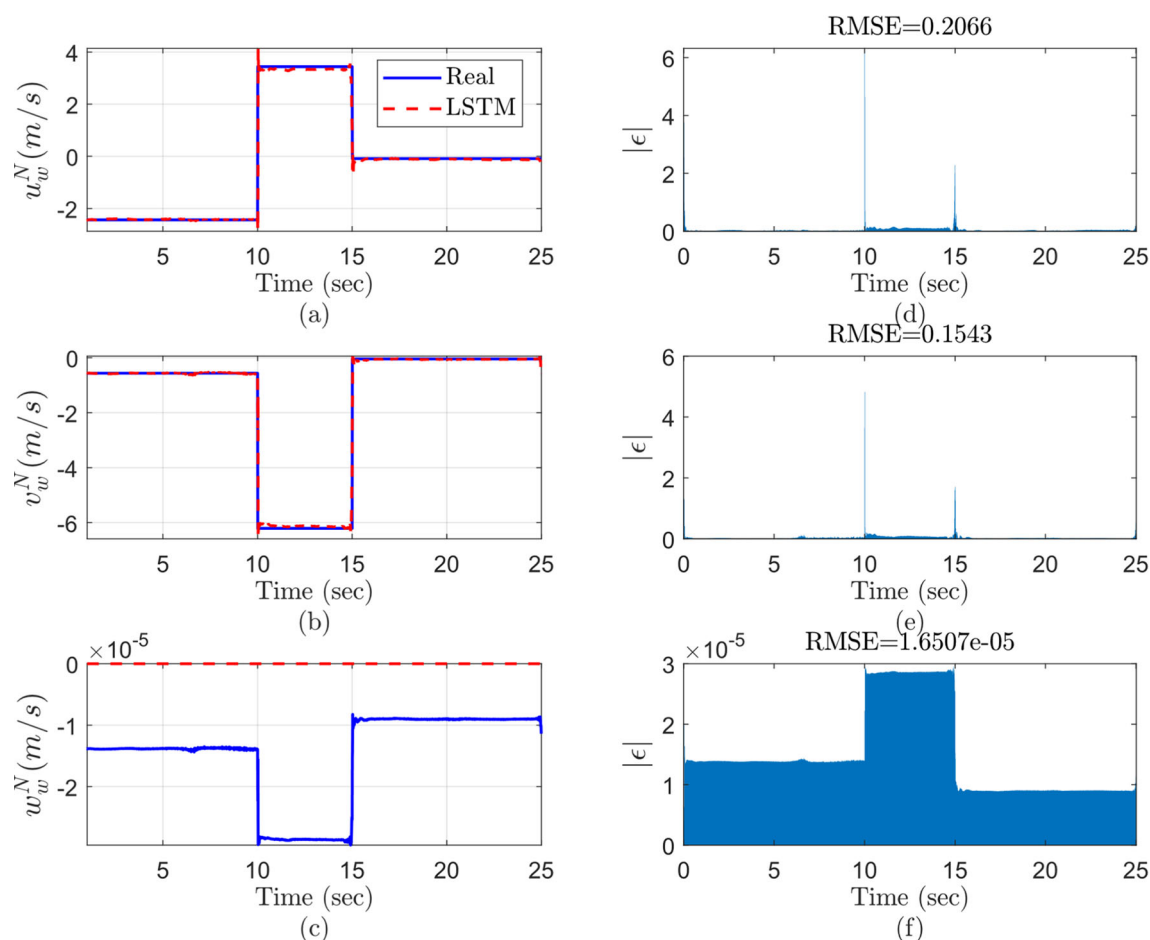


Fig. 11 Estimation of velocity of constant wind

with the real values. The precision of the estimation values is determined using error diagrams in second column of Figs. 16 and 17. These results indicated an acceptable tiny and bounded error in velocity estimation across the microburst wind profile. For microburst wind parameters, estimation errors converged to fixed values after around three seconds. Although the estimation values were not close to zero, they had a small percentage deviation in comparison to the real values. As an example, this deviation was less than four and two percent for r_{max} and z_{max} , respectively. The overall results indicated the trained network had acceptable accuracy after one and three seconds for estimation of wind velocity and wind parameters, respectively.

7 Verification and comparison of the results

The proposed algorithm has demonstrated significant accuracy in identifying and estimating the wind for sample data sets in Table 3 in the previous section. To further comprehensively evaluate the performance of this

algorithm, Monte Carlo (MC) simulation is performed in this section. Moreover, the algorithm is compared with a recent state-of-the-art wind estimation method.

7.1 Verification

To match the test results with real conditions, the parameters of the wind models were randomly selected in each MC simulation run between the range shown in Table 1. The MC test results were evaluated using confusion matrix (Japkowicz and Shah 2011) for wind model classification and error graphs for wind velocity vector and parameters estimation in the first and second layer of the proposed algorithm, respectively.

The confusion matrix is depicted by a matrix visualizing the performance of the proposed multi-class LSTM classifier on a set of test data. A data set with 100 members (25 members with different and randomly selected variables for each wind model) has been tested with the trained LSTM classifier. The confusion matrix is plotted in Fig. 17. The rows and columns of this matrix correspond to the predicted and true classes data, respectively, and label

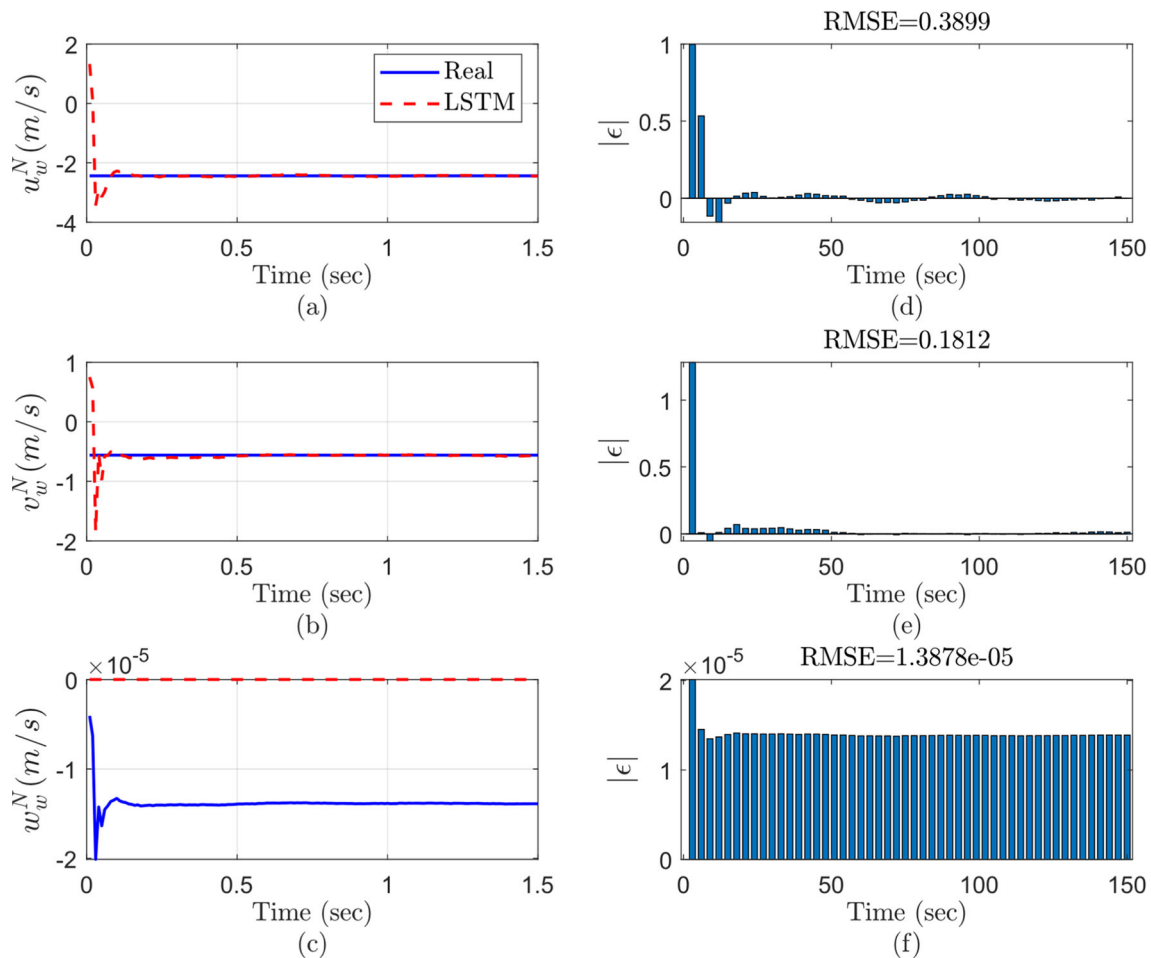


Fig. 12 The results for 1.5 s of constant wind in Fig. 11

numbers one to four indicate the type of wind model based on Table 1. The number and percentage of correctly and incorrectly classified wind models are shown in the diagonal and off-diagonal cells of the matrix, respectively. The column on the right and the row at the bottom of the confusion matrix show the percentages of the correctly and incorrectly classified of all predicted data. The results of Fig. 18 shows that the classifier could exactly identify constant wind, turbulence and 1-cosine wind with 100% accuracy, while for microburst wind accuracy was resulted to 96%. It is obvious that 4% of samples incorrectly classify in constant wind class. The overall accuracy equal to 99% is extracted and shown in the cell at the bottom right of the plot.

After the investigation of the specification of the incorrectly classified sample, it was found that the wind speed in all directions of this sample of microburst wind was close to zero. It is obvious this mistake is an acceptable margin in the practical application of the wind estimation problems. Generally, the high precision of these results demonstrated the ability and validity of trained

LSTM NN in the classification of the wind at the first layer of the proposed estimation algorithm.

The error histories of MC simulation (30 runs for each wind model) for constant wind, turbulence, 1-cosine, and microburst wind models are calculated and the mean and standard deviation of the error for each speed variable, are plotted in separate graphs. The mean (μ) and standard deviation (σ) of the error for constant wind, turbulence, 1-cosine, and microburst wind models are depicted in Figs. 20, 21, 22, and 23 in Appendix D, respectively.

The error histories show that, in some cases, the estimation errors do not converge to zero, but are bound to small values after a short time. Our investigation shows these bounded values are small enough for practical application and do not disturb the effectiveness of the proposed estimation algorithm. The estimation errors convergence bound values for the mean and standard deviation of the wind speed for all wind models are reported in Table 5.

Fig. 13 Estimation of velocity of turbulence (Dryden)

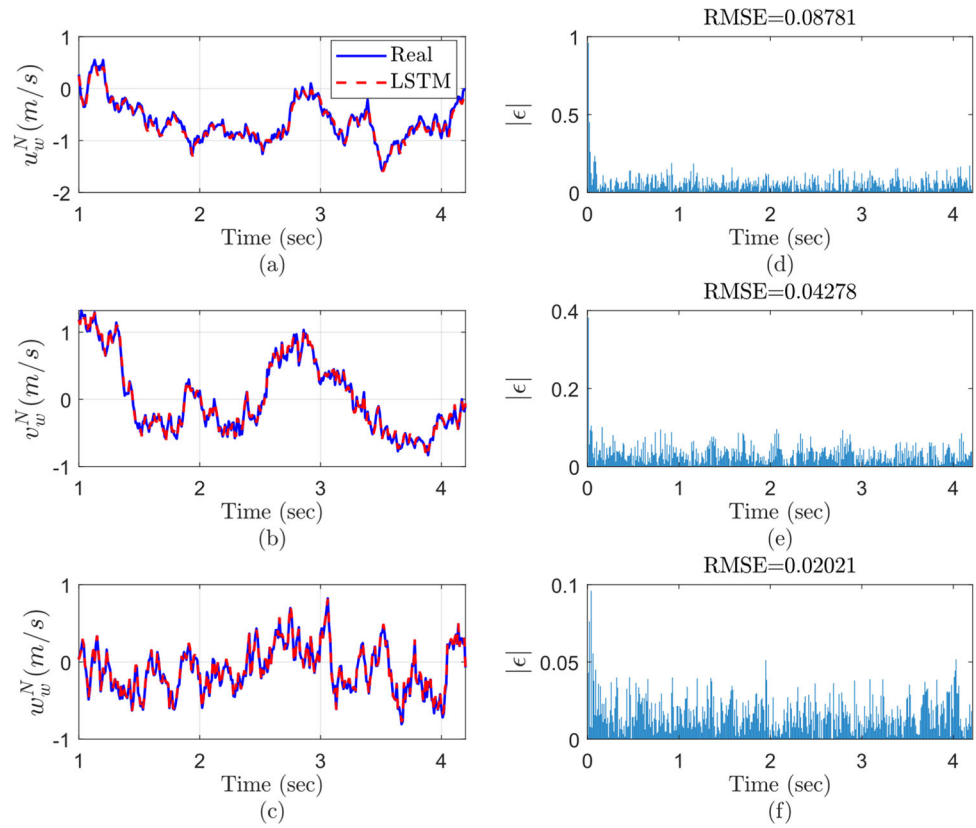


Fig. 14 Estimation of velocity of 1-cosine wind

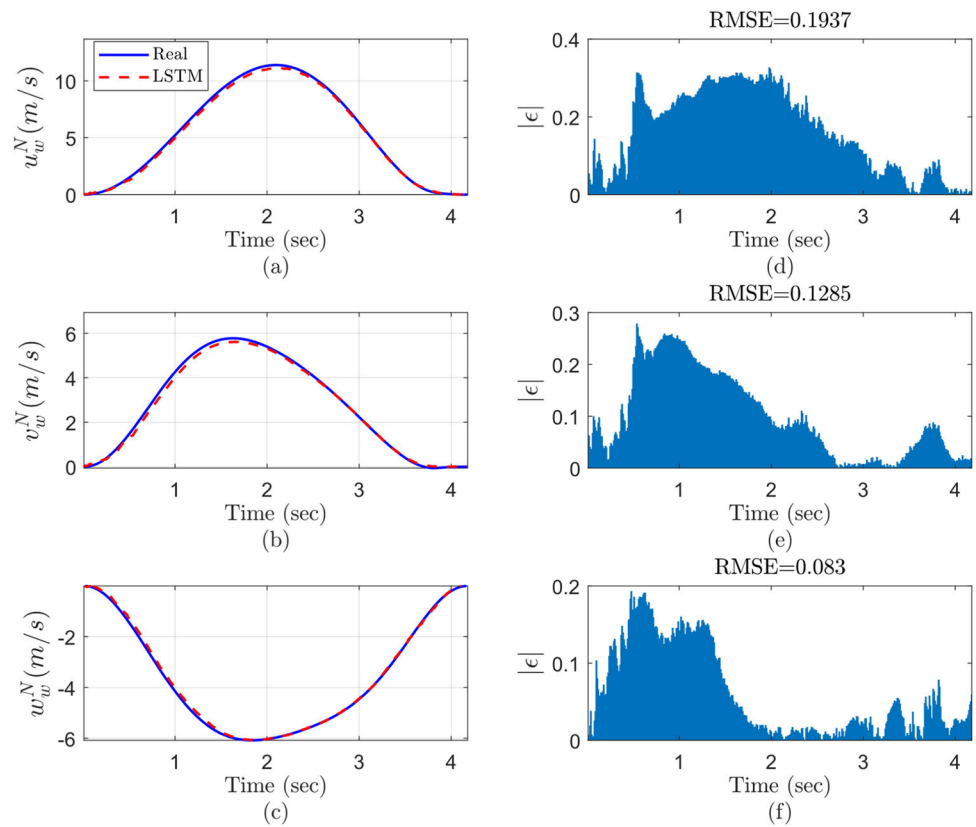
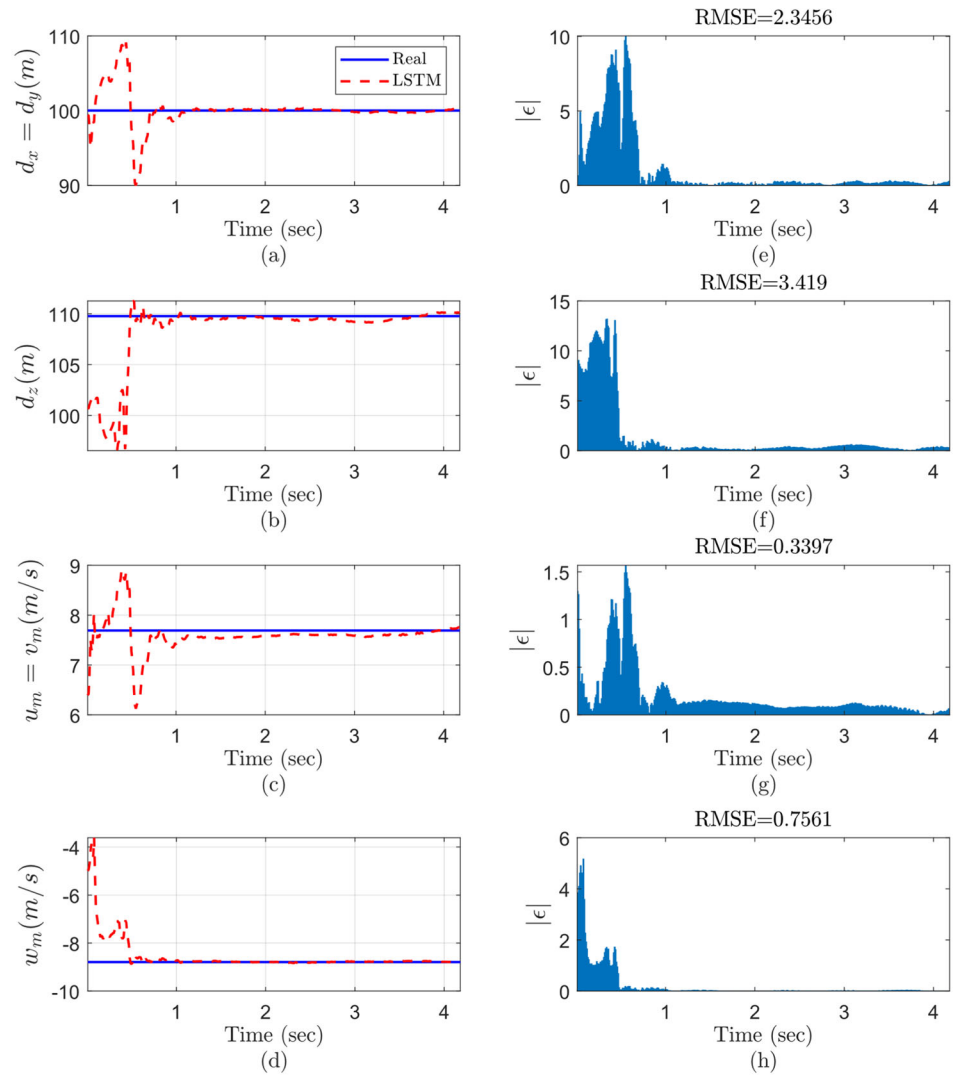


Fig. 15 Estimation of parameters of 1-cosine wind

7.2 Comparison

The proposed classification and estimation approaches in this work is based on a sequence of inputs including nine flight state variables of the UAV. The literature shows that conventional neural networks (CNN) are not suitable for working with this type of data (Liu et al. 2018; Medsker and Jain 2001), and cannot provide satisfactory results for our wind estimation problem. To compare the performance of our proposed wind estimation method with other neural networks, it is important to select an appropriate neural network that successfully handles sequence data. The Gated Recurrent Unit (GRU) (Cho et al. 2014), a type of recurrent neural network that is suitable for working with sequence data, can provide an appropriate comparison of the results. GRU is a simplified version of LSTM and only consists of two gates named reset and update. Information flow inside the GRU units is adjusted by the gating units without separating memory cells (Mirzaei et al. 2022). The

reset gate determines the amount of the previous hidden state to preserve, and the update gate determines the amount of new input that is incorporated into the hidden state. There is no separate output gate because the hidden state acts as both the cell state and output (Chen et al. 2019). To prepare a fair comparison between wind classification and estimation with LSTM and GRU, we used the same learning and test data, hyperparameters, architecture, and learning process with proposed LSTM NNs (described in Sect. 5). The comparison of LSTM and GRU in terms of training time indicates that in the same training conditions (including the same number of iterations and epochs), GRU needs less time for training because of its fewer parameters and simpler structure. In order to compare the performance of LSTM and GRU NNs, the criteria of accuracy and R-squared for classification of the wind models and estimation of wind speed and model parameters have been examined in in Tables 6 and 7, respectively.

Fig. 16 Estimation of velocity of microburst wind

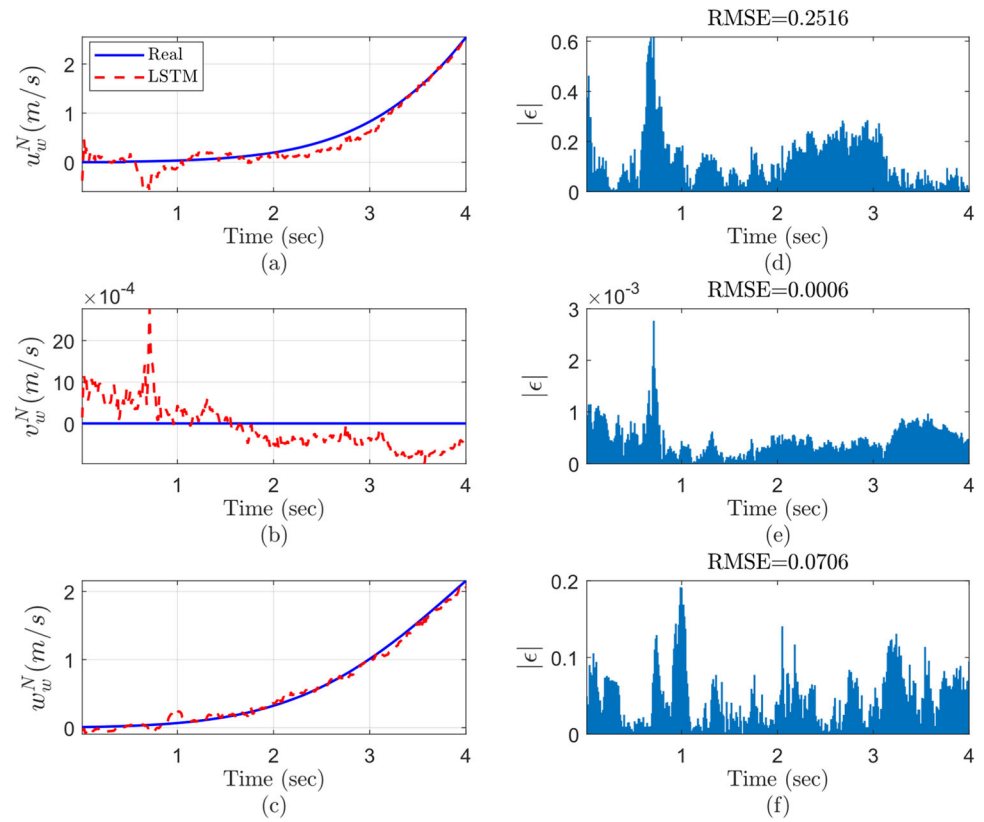
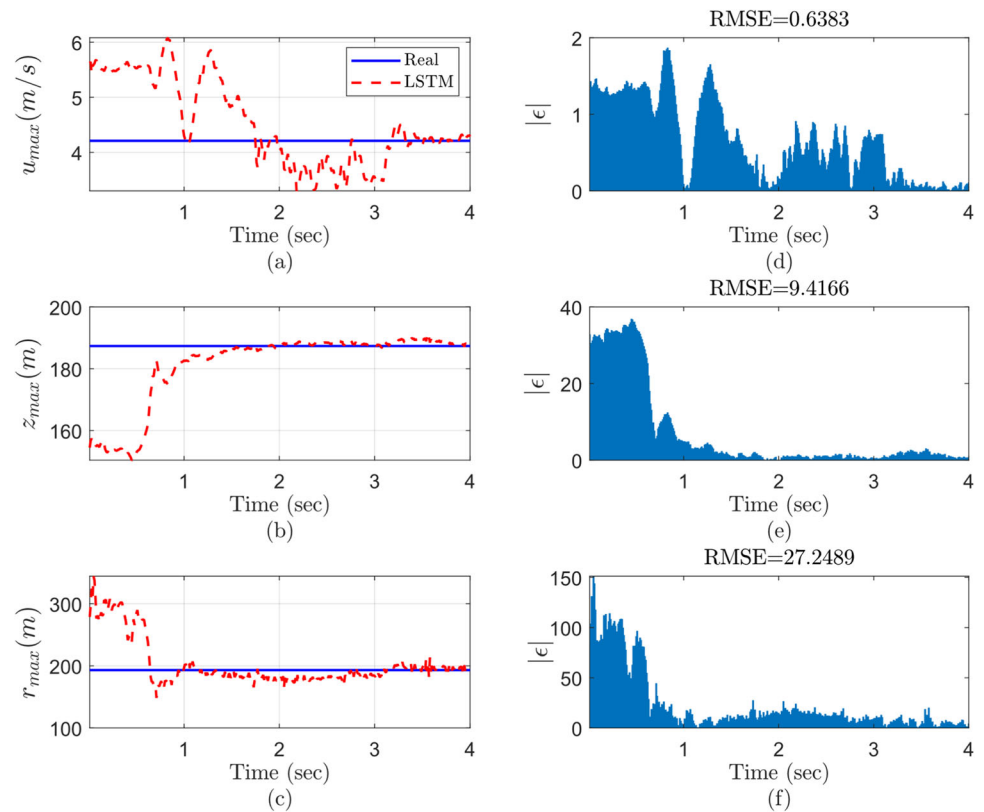


Fig. 17 Estimation of parameters of microburst wind



Confusion Matrix for Wind Classifier

Prediction Class	1	2	3	4	
	25 25.0%	0 0.0%	0 0.0%	1 1.0%	96.2% 3.8%
	0 0.0%	25 25.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	25 25.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	24 24.0%	100% 0.0%
Test Class					
1	100% 0.0%	100% 0.0%	100% 0.0%	96.0% 4.0%	99.0% 1.0%
2					
3					
4					

Fig. 18 Confusion matrix for trained LSTM NN multi-class classifier

To present reliable statistical analysis for the classification of wind models, 30 confusion matrixes are plotted for 30 data sets. The mean and standard deviation of the accuracy of the LSTM and GRU NNs for classified wind models are shown in Table 6. The results show that under the same learning and testing conditions, LSTM is more accurate than GRU in recognizing wind models.

To evaluate the effectiveness of the proposed wind estimation LSTM NNs on all test data, we utilized the

coefficient of determination or R-squared (R^2) criterion. It measures the accuracy of the observed outcomes replicated by the proposed models. In order to calculate R-squared values, we have used Eq. (39) and random samples in the Monte-Carlo simulation, so that the last test and estimated values in the investigated time frame have been used for each model. The R-squared score normally ranges from zero to one, where larger values of the R-squared score indicate better accuracy of the developed NNs. Table 7 presents the measured R-squared for estimated velocities and parameters of the different wind models with LSTM and GRU NNs. The results show significant superiority of LSTM NNs over GRU NNs in estimating the wind speed and parameters of various wind models.

The wind models, UAV configuration, and flight condition investigated in this paper are nearly identical to those used in (Nobahari and Sharifi 2020). Therefore, the results obtained in this study are compared with those obtained from the state-of-the-art estimation method described in (Nobahari and Sharifi 2020) by H. Nobahari et al. The wind estimation method presented by H. Nobahari et al. used a heuristic multiple model filter based on the outputs that provided by three rate gyros, an altimeter, a digital compass and two pitot in a fixed-wing UAV. The presented results in this reference graphically illustrate the great performance of the Extended Kalman Filter (EKF) in the wind estimating problem. The convergence time of estimation is only the quantity information that is extractable from the presented graphs. They show an estimation with almost 100% accuracy (RMSE = 0) after a few seconds (A page limit prevents us from presenting them here). The convergence time for the presented results

Table 5 The estimation errors convergence bound values for wind velocities estimation results

Wind model	Parameter	Convergence bound		Wind model	Parameter	Convergence bound	
		μ	σ			μ	σ
Constant	u_g	0.03	0.02	Dryden	u_g	0.20	0.19
	v_g	0.03	0.02		v_g	0.19	0.16
	w_g	0.00	0.00001		w_g	0.09	0.11
1-cosine	u_g	0.25	0.82	Microburst	u_g	0.23	0.16
	v_g	0.25	0.04		v_g	0.0	0.0001
	w_g	0.10	0.042		w_g	0.07	0.05

Table 6 The accuracy of the LSTM and GRU NNs for classification of wind models

Wind models	Constant wind		Dryden wind		1-cosine		Microburst	
	μ	σ	μ	σ	μ	σ	μ	σ
LSTM	0.98	0.02	0.98	0.01	0.97	0.01	0.94	0.03
GRU	0.94	0.01	0.93	0.02	0.88	0.03	0.87	0.04

Table 7 The R-squared measure for the estimated velocities and parameters of the different wind

Wind model	Parameter	R^2 criterion		Wind model	Parameter	R^2 criterion	
		LSTM	GRU			LSTM	GRU
Constant	u_w	0.990	0.750	Dryden	u_w	0.951	0.760
	v_w	0.990	0.750		v_w	0.982	0.780
	w_w	1.000	0.800		w_w	0.961	0.710
1-cosine	u_w	0.975	0.750	Microburst	u_w	0.984	0.650
	v_w	0.981	0.850		v_w	0.978	0.620
	w_w	0.992	0.820		w_w	0.981	0.630
	$u_m = v_m$	0.973	0.760		u_{max}	0.973	0.610
	w_m	0.978	0.680		z_{max}	0.922	0.590
	$d_x = d_y$	0.985	0.790		r_{max}	0.936	0.650
	d_z	0.982	0.780		—	—	—

Table 8 The comparison of the results of LSTM with extended Kalman filter estimation method

Wind model	Parameter	Estimation time (sec)		Wind model	Parameter	Estimation time (sec)	
		LSTM	EKF			LSTM	EKF
Constant	u_w	0.7	1.2	Dryden	u_w, v_w, w_w	0.5	—
	v_w	0.7	—		V_{wh6}	0.5	—
	w_w	$\cong 0$	1.2		ψ_w	0.5	—
1-cosine	u_w	< 1	1.4	Microburst	u_g	< 1	2
	v_w	< 1	—		v_g	$\cong 0$	—
	w_w	< 1	1.4		w_g	< 1	2
	$u_m = v_m$	1.2	1.2		u_{max}	3.5	—
	w_m	1	1.2		z_{max}	4	—
	$d_x = d_y, d_z$	1.2	0		r_{max}	3.5	4

in (Nobahari and Sharifi 2020) and our proposed algorithm is compared in Table 8. Also, the information presented in Table 4 showed the RMSE of results obtained based on our estimation algorithm. Although the displayed results in some cases indicate RMSE greater than zero, these values are within an acceptable range.

The estimation results show that the Kalman filters converged estimation values almost after more than one second, while the coverage time for identifying the type of wind and estimating wind velocity is less than one second for our proposed algorithm. In addition, the results indicated our algorithm had a shorter estimation time for other wind models. Although our proposed algorithm provides less accuracy compared to the EKF, it is superior for estimating various types of wind because of its faster estimation time and direct use of sensor outputs without relying on the air data system and the navigation system.

8 Conclusions

This work proposes a two-step LSTM NN method to identify wind models, estimate wind parameters, and wind velocity vector using only AHRS measurement data. One classification and four regression LSTM NNs are trained using data from nonlinear 6DOF simulations of the motion of a small fixed-wing UAV. The proposed algorithm and networks were analyzed in detail for different wind models through Monte Carlo simulation. The results showed that the appropriate training networks using a sequence of simple measurement data could first identify the wind model and second estimate the wind velocity component in less than one second also estimate the wind model parameters faster than another new wind estimation method. The comparison of LSTM and GRU reveals that the LSTM has a higher accuracy in identifying wind models and estimating wind parameters and velocity vector.

The proposed approach directly uses the outputs of the onboard sensors and does not rely on the navigation and air data systems of the UAV with low-frequency response. As a result, this algorithm has a high estimation speed, making it suitable for real-time wind estimation applications. This wind estimation method is applicable in dynamic flight phases and unsteady conditions that occur in the presence of turbulent winds. Furthermore, it is a practical method for determining wind in areas where GPS signals are weak or denied or meteorology measuring equipment is unavailable. Applying this method in commercial UAVs can lead to removing the air data system and reducing their weight and price. Additionally, the proposed algorithm could serve as an emergency backup for large or small aircraft whose air data sensors fail by training the networks for all likely aircraft maneuvers. It can be a low-cost and accurate wind estimation technique to apply to any general fixed-wing aircraft without requiring extra instrumentation, GPS signals, knowledge of the controller, or dynamic models of the UAV.

A future interesting research line would be implementing this algorithm on a real fixed-wing UAV in a real-world scenario. The satisfactory statistical results obtained here can promise the successful testing of this algorithm in real scenarios. Furthermore, it would be interesting and more applicable to extend this work to consider a combination of the different type of winds. This extension will need a new simulation of the flight in the simultaneous presence of multi wind models and training new LSTM networks. The use of optimization methods to choose hyperparameters of the proposed NNs would be another remarkable piece of future work.

Appendix A: Equations of motion of a fixed-wing UAV with external disturbances

The translational and rotational six-degree of freedom equations of motion of the UAV subjected to the aerodynamic and thrust forces $\mathbf{f}_{a,p}$ and the gravitational acceleration $\mathbf{g}$ in the body frame (by assuming flat non-rotating earth for an Aircraft with low-speed and short flight range) shown in Fig. 19 can be written as (Zipfel 2014):

$$m \left[\frac{d\mathbf{v}_B^E}{dt} \right]^B = [\mathbf{f}_{a,p}]^B + m[\mathbf{g}]^B - m[\boldsymbol{\Omega}^{BE}]^B [\mathbf{v}_B^E]^B \quad (\text{A.1})$$

$$\left[\frac{d\mathbf{s}_{BE}}{dt} \right]^L = [\mathbf{R}]^{BL} [\mathbf{v}_B^E]^B \quad (\text{A.2})$$

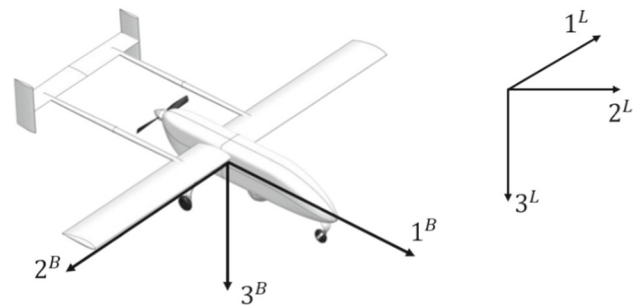


Fig. 19 Body frame definition relative to a local frame

$$\left[\frac{d\boldsymbol{\omega}^{BE}}{dt} \right]^B = \left([\mathbf{I}_B^B]^B \right)^{-1} \left(-[\boldsymbol{\Omega}^{BE}]^B [\mathbf{I}_B^B]^B [\boldsymbol{\omega}^{BE}]^B + [\mathbf{m}_B]^B \right) \quad (\text{A.3})$$

where $[\mathbf{v}_B^E]^B = [u \ v \ w]^T$ is the linear velocity of UAV w.r.t Earth, $[\mathbf{s}_{BE}]^L = [x \ y \ z]^T$ is the position of the UAV in the local frame, $[\boldsymbol{\Omega}^{BE}]^B$ is the skew-symmetric form of the angular velocity of the UAV w.r.t Earth $[\boldsymbol{\omega}^{BE}]^B = [p \ q \ r]^T$, $[\mathbf{f}_{a,p}]^B$ and $[\mathbf{m}_B]^B$ are the aerodynamic and propulsive forces and moments expressed in the body frame, respectively. Also, m represents the mass of the UAV and $[\mathbf{I}_B^B]^B$ is the inertia matrix in body frame that expressed as:

$$[\mathbf{I}_B^B]^B = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{xz} & 0 & I_{zz} \end{bmatrix}^B \quad (\text{A.4})$$

It is possible to write the gravitational acceleration $[\mathbf{g}]^B$ that expressed in body frame in a local-level coordinate system with a transformation matrix $[\mathbf{R}]^{BL}$ as $[\mathbf{g}]^B = [\mathbf{R}]^{BL} [\mathbf{g}]^L$. This transformation matrix using the quaternion relationship is expressed as:

$$[\mathbf{R}]^{BL} = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 + q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix} \quad (\text{A.5})$$

The four-quaternion elements q_0 , q_1 , q_2 , and q_3 are related to the three body-rate components p , q and r , by four linear differential equations as follows:

$$\begin{bmatrix} \dot{q}_0 \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix} \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} \quad (\text{A.6})$$

The gravitational acceleration in the local frame can be expressed as:

$$[\mathbf{g}]^L = [0 \ 0 \ g]^T \quad (\text{A.7})$$

Furthermore, $[\mathbf{f}_{a,p}]^B$ is the sum of the aerodynamic and propulsive forces acting on the airframe.

$$[\mathbf{f}_{a,p}]^B = [\mathbf{f}_a]^B + [\mathbf{f}_p]^B \quad (\text{A.8})$$

With Assuming that the trust is symmetric and is aligned with the longitudinal body axes, and with defining T and δ_t as the total thrust and the throttle setting angle, respectively, the propulsive force in the body frame is assumed as:

$$[\mathbf{f}_p]^B = [T\delta_t \ 0 \ 0]^T \quad (\text{A.9})$$

While the aerodynamic force and moment coefficients $C_X, C_Y, C_Z, C_l, C_m, C_n$ defined in the body coordinate system with a positive direction along the body axes, the aerodynamic force and moment vectors $\mathbf{f}_a^B = [f_X \ f_Y \ f_Z]^T$ and $\mathbf{m}^B = [l \ m \ n]^T$ expressed in the body frame as (Zipfel 2014):

$$[\mathbf{f}_a]^B = \frac{1}{2} \rho V_a^2 S [C_X \ C_Y \ C_Z]^T \quad (\text{A.10})$$

$$[\mathbf{m}_B]^B = \frac{1}{2} \rho V_a^2 S [C_l \times b \ C_m \times \bar{c} \ C_n \times b]^T \quad (\text{A.11})$$

where ρ is the air density, S is the reference (wing) area, b is the wing span and $\bar{c}$ is the mean aerodynamic chord, respectively. Also V_a is the airspeed, by defining the relative velocity of the UAV w.r.t the wind, $\mathbf{v}_a^B = [u_{rel} \ v_{rel} \ w_{rel}]^T$, it calculated as:

$$V_a = \|\mathbf{v}_a^B\| = \sqrt{u_{rel}^2 + v_{rel}^2 + w_{rel}^2} \quad (\text{A.12})$$

The aerodynamic force and moment coefficients are defined as:

$$\begin{bmatrix} C_X \\ C_Y \\ C_Z \end{bmatrix} = \begin{bmatrix} C_{X_0} + C_{X_\alpha} \alpha + C_{X_\dot{\alpha}} \dot{\alpha} + C_{X_q} q_{rel} \frac{\bar{c}}{2V_a} + C_{X_{\delta_e}} \delta_e \\ C_{Y_0} + C_{Y_\beta} \beta + (C_{Y_p} p_{rel} + C_{Y_r} r_{rel}) \frac{b}{2V_a} + C_{Y_{\delta_a}} \delta_a + C_{Y_{\delta_r}} \delta_r \\ C_{Z_0} + C_{Z_\alpha} \alpha + C_{Z_\dot{\alpha}} \dot{\alpha} + C_{Z_q} q_{rel} \frac{\bar{c}}{2V_a} + C_{Z_{\delta_e}} \delta_e \end{bmatrix} \quad (\text{A.13})$$

$$\begin{bmatrix} C_l \\ C_m \\ C_n \end{bmatrix} = \begin{bmatrix} C_{l_0} + C_{l_\beta} \beta + (C_{l_p} p_{rel} + C_{l_r} r_{rel}) \frac{b}{2V_a} + C_{l_{\delta_a}} \delta_a + C_{l_{\delta_r}} \delta_r \\ C_{m_0} + C_{m_\alpha} \alpha + C_{m_\dot{\alpha}} \dot{\alpha} + C_{m_q} q_{rel} \frac{\bar{c}}{2V_a} + C_{m_{\delta_e}} \delta_e \\ C_{n_0} + C_{n_\beta} \beta + (C_{n_p} p_{rel} + C_{n_r} r_{rel}) \frac{b}{2V_a} + C_{n_{\delta_a}} \delta_a + C_{n_{\delta_r}} \delta_r \end{bmatrix} \quad (\text{A.14})$$

By defined C_L and C_D as lift and drag coefficients, the non-dimensional longitudinal aerodynamic force coefficients C_X and C_Z expressed as:

$$\begin{aligned} C_{X_i} &= -C_{D_i} \cos \alpha + C_{L_i} \sin \alpha \quad \forall i = 0, \alpha, \dot{\alpha}, q, \delta_e \\ C_{Z_i} &= -C_{D_i} \sin \alpha - C_{L_i} \cos \alpha \quad \forall i = 0, \alpha, \dot{\alpha}, q, \delta_e \end{aligned} \quad (\text{A.15})$$

In the above equations, $\boldsymbol{\omega}_a^B = [p_{rel} \ q_{rel} \ r_{rel}]^T$ defined as angular velocity of the UAV relative to the air. Also, the parameters α and β are the angle of attack and sideslip, respectively, and calculated as:

$$\alpha = \tan^{-1} \frac{w_{rel}}{u_{rel}} \quad (\text{A.16})$$

$$\beta = \sin^{-1} \frac{v_{rel}}{V_a} \quad (\text{A.17})$$

By defined $\mathbf{v}_w^L = [u_w^L \ v_w^L \ w_w^L]^T$ and $\boldsymbol{\omega}_w^L = [p_w^L \ q_w^L \ r_w^L]^T$ as linear and angular velocity of wind in local frame, and $\mathbf{v}^B = [u \ v \ w]^T$, $\boldsymbol{\omega}_{LB}^B = [p \ q \ r]^T$ as linear velocity of UAV in body and angular velocity of the body frame with respect to the local frame, expressed in the body frame, respectively, the $\mathbf{v}_a^B$ and $\boldsymbol{\omega}_a^B$ calculated as:

$$\mathbf{v}_a^B = \mathbf{v}^B - [\mathbf{R}]^{BL} \mathbf{v}_w^L = \mathbf{v}^B - \mathbf{v}_w^B \quad (\text{A.18})$$

$$\boldsymbol{\omega}_a^B = \boldsymbol{\omega}^B - [\mathbf{R}]^{BL} \boldsymbol{\omega}_w^L = \boldsymbol{\omega}^B - \boldsymbol{\omega}_w^B \quad (\text{A.19})$$

where $\mathbf{v}_w^B = [u_w \ v_w \ w_w]^T$ and $\boldsymbol{\omega}_w^B = [p_w \ q_w \ r_w]^T$ are wind velocity and angular velocity of wind in body frame, respectively. The Euler angles ϕ , θ and ψ can be calculated as:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (\text{A.20})$$

Appendix B: Autopilot specification and parameters of the simulated UAV

The specifications of acceleration autopilot for the developed closed-loop nonlinear six-degree of freedom simulator are given in Table 9. This information includes inputs, outputs, and control gains for the normal, lateral and bank acceleration control system of a fixed-wing UAV in the form of Fig. 8 (Esmailifar and Saghafi 2017).

The geometry, inertia and aerodynamic parameters of a small fixed-wing UAV used as a case study in this paper are presented in Tables 10 and 11, respectively. The aerodynamic parameters include dimensionless aerodynamic coefficients that are used in the closed-loop nonlinear six-degree of freedom simulation and guidance and control system design process.

Table 9 The specification of the autopilot (Esmailifar and Saghaei 2017)

Channel	Input and output	Control gains
Normal Acc. Autopilot	$X = \begin{bmatrix} q \\ az \end{bmatrix}$ and $x_c = az_c$	$K_1 = \frac{2\omega_{nsp} \xi_{sp} + p_{sp} + M_q + (Z_x/V)}{M_{\delta_e}}$ $K_2 = -\frac{2\omega_{nsp} \xi_{sp} p_{sp} + \omega_{nsp}^2 + M_z + (Z_x/V)(k_1 M_{\delta_e} - M_q)}{Z_x M_{\delta_e}}$ $K_I = \frac{\omega_{nsp}^2 p_{sp}}{Z_x M_{\delta_e}}$
Lateral Acc. Autopilot	$X = \begin{bmatrix} r \\ a_Y \end{bmatrix}$ and $x_c = a_{Y_c}$	$K_1 = \frac{2\omega_{nl} \xi_L + p_L + N_r + (Y_\beta/V)}{N_{\delta_r}}$ $K_2 = \frac{2\omega_{nl} \xi_L p_L + \omega_{nl}^2 + N_\beta + (Y_\beta/V)(k_1 N_{\delta_r} - N_r)}{Y_\beta N_{\delta_r}}$ $K_I = -\frac{\omega_{nl}^2 p_L}{Y_\beta N_{\delta_r}}$
Bank Angle Autopilot	$X = \begin{bmatrix} p \\ \phi \end{bmatrix}$ and $x_c = \phi_c$	$K_1 = \frac{2\omega_{nR} \xi_R + p_R + LL_p}{LL_{\delta_a}}$ $K_2 = \frac{2\omega_{nR} \xi_R p_R + \omega_{nR}^2}{LL_{\delta_a}}$ $K_I = -\frac{\omega_{nR}^2 p_R}{LL_{\delta_a}}$

$$M_q = \frac{\bar{q} S c^2}{2v I_{yy}} C_{m_q}, M_{\delta_e} = \frac{\bar{q} S c}{I_{yy}} C_{m_{\delta_e}}, Z_q = -\frac{\bar{q} S}{m} C_{L_z}$$

$$N_\beta = \frac{\bar{q} S b}{I_{zz}} C_{n_\beta}, N_r = \frac{\bar{q} S b^2}{2v I_{zz}} C_{n_r}, N_{\delta_r} = \frac{\bar{q} S b}{I_{zz}} C_{n_{\delta_r}}, Y_\beta = -\frac{\bar{q} S}{m} C_{Y_\beta}, LL_p = \frac{\bar{q} S b^2}{2v I_{xx}} C_{l_p}, LL_{\delta_a} = \frac{\bar{q} S b}{I_{xx}} C_{l_{\delta_a}}$$
Table 10 The UAV geometry and inertia parameters

Parameter	Value	Unit	Parameter	Value	Unit
S	0.55	m ²	I_{xx}	0.82	kg.m ²
b	2.9	m	I_{yy}	1.14	kg.m ²
$\bar{c}$	0.19	m	I_{zz}	1.76	kg.m ²
m	13.5	kg	I_{xz}	0.12	kg.m ²

Table 11 The UAV longitudinal and lateral-directional stability and control coefficients

Parameter	Value	Parameter	Value	Parameter	Value
C_{L_0}	0.230	$C_{m_{\delta_e}}$	- 0.99	C_{l_p}	- 0.505
C_{Y_0}	0.0	C_{Y_β}	- 0.83	C_{n_p}	- 0.069
C_{D_0}	0.043	C_{l_β}	- 0.13	C_{Y_r}	0.0
C_{l_0}	0.0	C_{n_β}	0.073	C_{l_r}	0.252
C_{m_0}	0.135	C_{L_z}	1.97	C_{n_r}	- 0.095
C_{n_0}	0.0	C_{D_z}	0.0	$C_{Y_{\delta_r}}$	0.191
C_{L_x}	5.61	C_{m_z}	- 10.38	$C_{l_{\delta_r}}$	0.0024
C_{D_x}	0.149	C_{L_q}	7.95	$C_{n_{\delta_r}}$	- 0.069
C_{m_x}	- 2.74	C_{D_q}	0.0	$C_{Y_{\delta_a}}$	- 0.075
$C_{L_{\delta_e}}$	0.13	C_{m_q}	- 38.21	$C_{l_{\delta_a}}$	- 0.169
$C_{D_{\delta_e}}$	0.036	C_{Y_p}	0.0	$C_{n_{\delta_a}}$	0.0108

Appendix C: The selection process of the LSTM's hyperparameters

Hyperparameters as the external configuration variables of the LSTM NNs has been tuned after a series of manual trials on test data. Table 12, 13, 14, 15 and 16 present the history of tuning process for the proposed LSTM NNs. This information indicates that the trial process for setting hyperparameters of the wind classification NN and regression NNs of the constant and Dryden wind models has continued until achieve to the desired accuracy or RMSE. Also, they show that the hyperparameters tuning process has terminated when changing hyperparameters did not affect the RMSE criteria for regression NNs of the 1-cosine and microburst wind models.

Table 12 The history of setting hyperparameters of the classification LSTM NN

Hidden unit	50	50	75	100	100	100	100	100
Min batch size	10	15	15	20	20	30	25	25
Epochs of training	50	50	50	100	150	150	200	250
Accuracy (%)	50	50	60	70	85	80	99	99

Table 13 The history of setting hyperparameters of the regression NN for constant wind

Hidden unit	50	100	100	150	200	200	200	200
Min batch size	10	40	30	30	30	20	30	30
Epochs of training	50	100	100	100	150	250	300	350
RMSE (end)	1.5	0.2	0.15	0.1	0.05	0.01	0.008	0.008

Table 14 The history of setting hyperparameters of the regression NN for Dryden wind

Hidden unit	50	50	100	150	200	200	200	200
Min batch size	10	30	20	25	30	27	27	27
Epochs of training	50	100	100	100	150	200	250	300
RMSE (end)	1	0.8	0.2	0.1	0.07	0.01	0.002	0.002

Table 15 The history of setting hyperparameters of the regression NN for 1-cosine wind

Hidden unit	50	100	100	150	200	200	200	200
Min batch size	10	40	20	25	30	28	27	27
Epochs of training	50	50	100	100	150	150	200	250
RMSE (end)	2	0.5	0.2	0.18	0.8	0.03	0.01	0.01

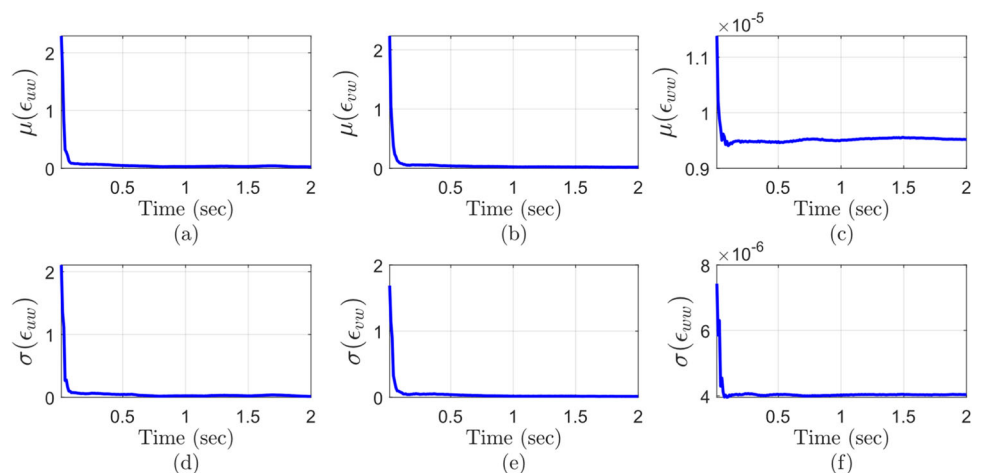
Table 16 The history of setting hyperparameters of the regression NN for microburst wind

Hidden unit	50	50	100	150	200	200	200	200
Min batch size	10	20	30	30	20	25	25	25
Epochs of training	50	50	50	70	100	100	150	200
RMSE (end)	5	4	1.5	0.8	0.5	0.25	0.2	0.2

Appendix D: Monte Carlo simulation results

The mean and standard deviation history of the Monte Carlo test results for different wind models are presented in Figs. 20, 21, 22 and 23.

Fig. 20 Mean and standard deviation of the Monte Carlo test results for constant wind estimation error in each time step: **a** $\mu_{\epsilon_{uw}}$, **b** $\mu_{\epsilon_{vw}}$, **c** $\mu_{\epsilon_{ww}}$, **d** $\sigma_{\epsilon_{uw}}$, **e** $\sigma_{\epsilon_{vw}}$, **f** $\sigma_{\epsilon_{ww}}$



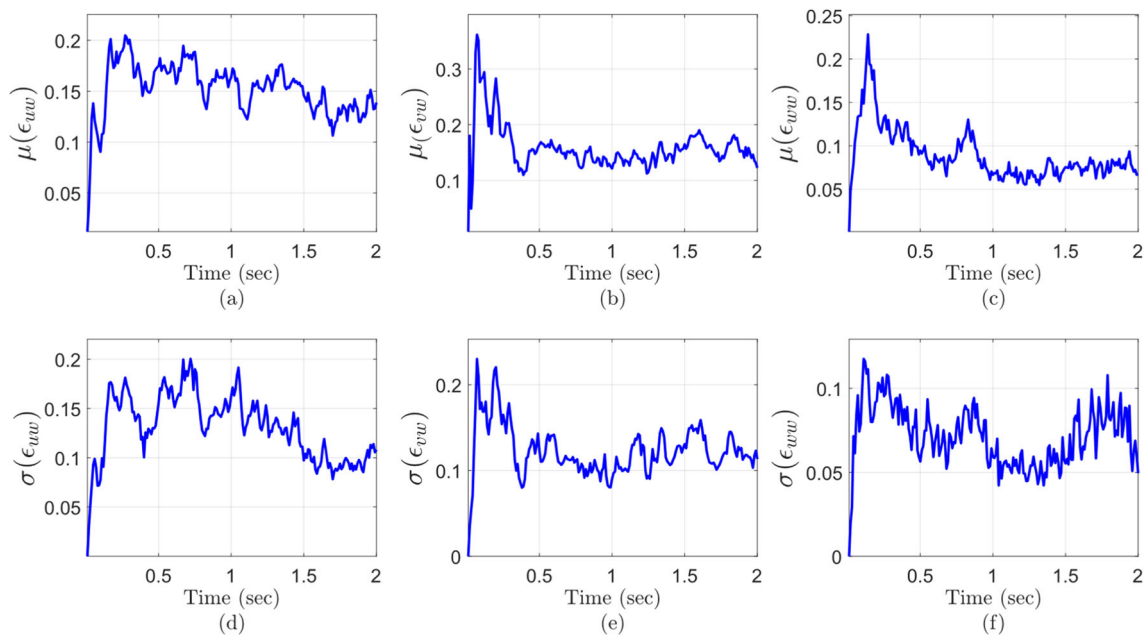
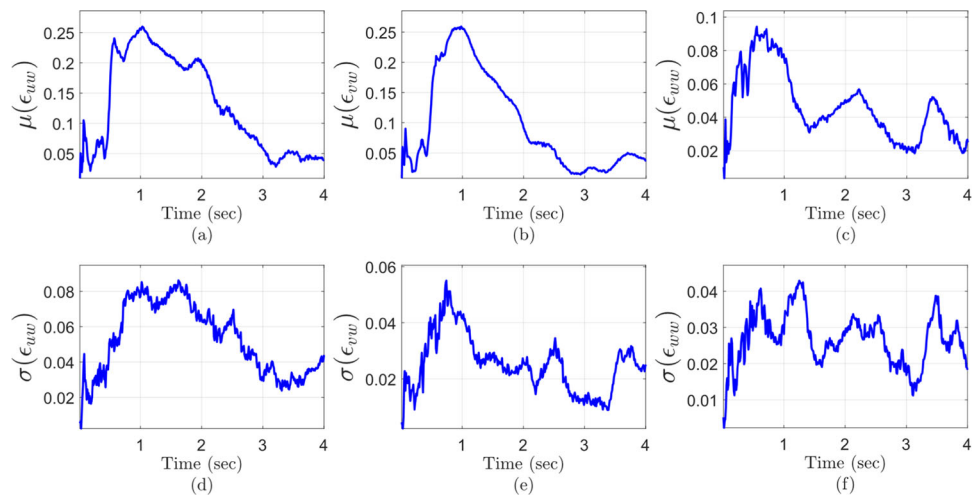


Fig. 21 Mean and standard deviation of the Monte Carlo test results for turbulence (Dryden) estimation error in each time step: **a** $\mu_{\epsilon_{uw}}$, **b** $\mu_{\epsilon_{vw}}$, **c** $\mu_{\epsilon_{wv}}$, **d** $\sigma_{\epsilon_{uw}}$, **e** $\sigma_{\epsilon_{vw}}$, **f** $\sigma_{\epsilon_{wv}}$

Fig. 22 Mean and standard deviation of the Monte Carlo test results for 1-cosine wind estimation error in each time step: **a** $\mu_{\epsilon_{uw}}$, **b** $\mu_{\epsilon_{vw}}$, **c** $\mu_{\epsilon_{wv}}$, **d** $\sigma_{\epsilon_{uw}}$, **e** $\sigma_{\epsilon_{vw}}$, **f** $\sigma_{\epsilon_{wv}}$



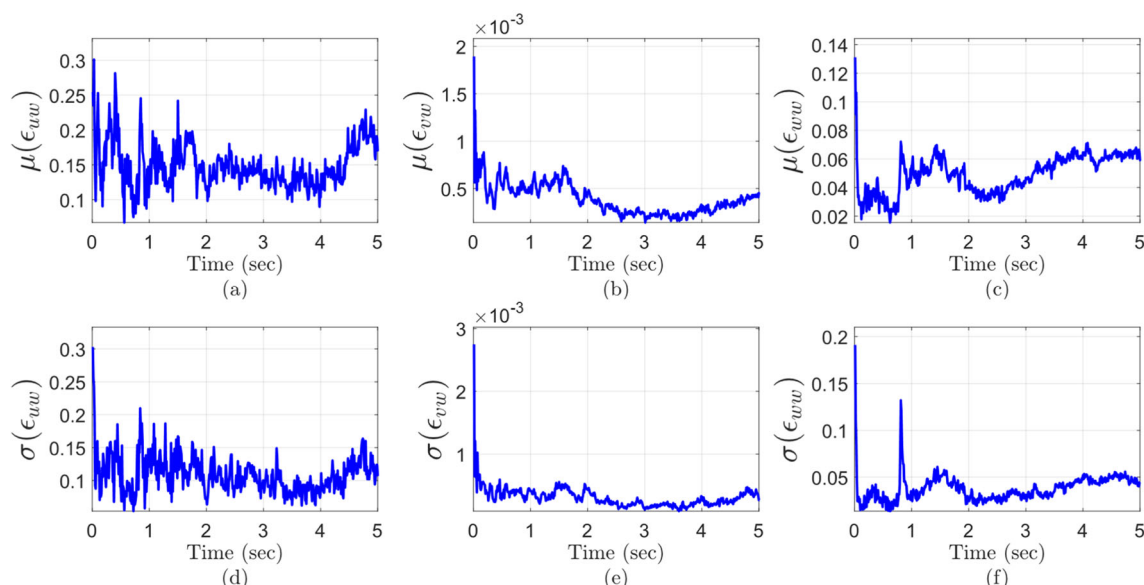


Fig. 23 Mean and standard deviation of the Monte Carlo test results for microburst wind estimation error in each time step: **a** $\mu_{\epsilon_{uw}}$, **b** $\mu_{\epsilon_{vw}}$, **c** $\mu_{\epsilon_{wv}}$, **d** $\sigma_{\epsilon_{uw}}$, **e** $\sigma_{\epsilon_{vw}}$, **f** $\sigma_{\epsilon_{wv}}$

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Data availability The datasets generated during the current study are not publicly available due to it is owned by Amirkabir University of Technology but are available from the corresponding author on reasonable request.

Declarations

Conflict of interest All authors state that there is no conflict of interest.

Ethical approval We used our own data. Animals and Humans are not involved in this research work.

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